



Robust Machine Learning for Real-world Time-series Applications

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October 8, 2025



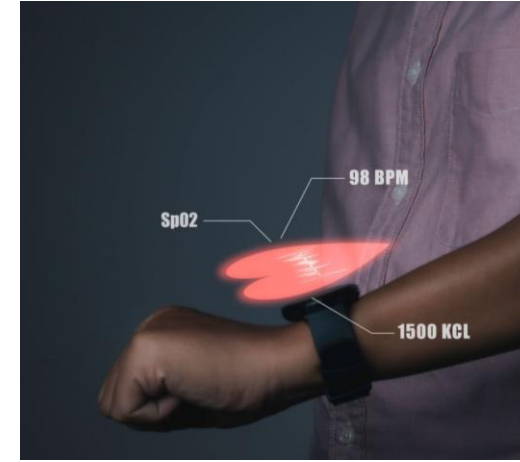
Time Series Data are Ubiquitous



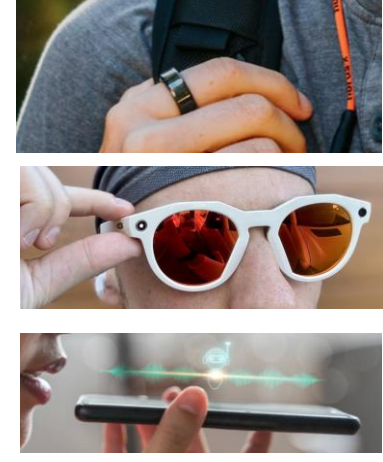
Clinical Applications



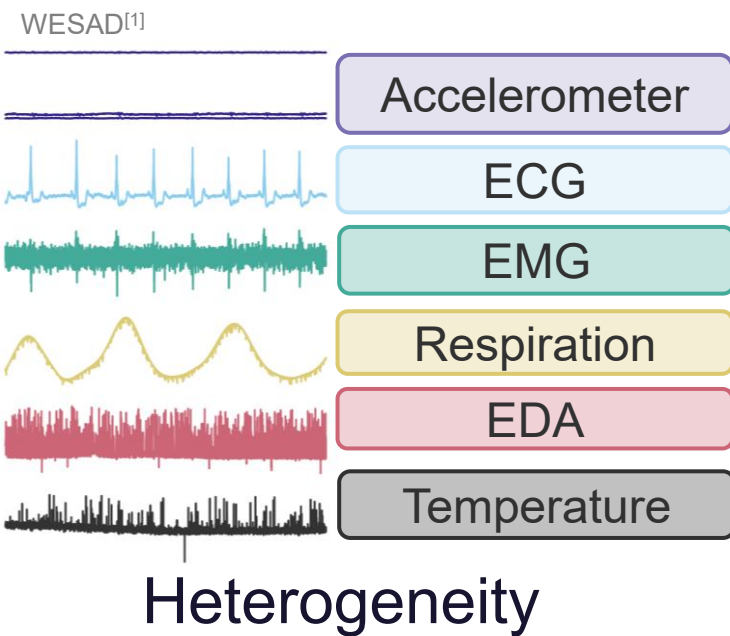
Environment Monitoring



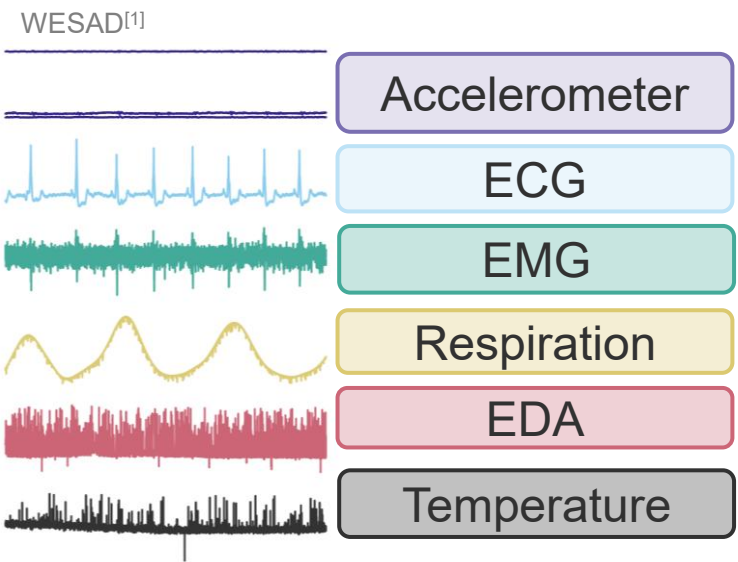
Lifestyle Enhancement



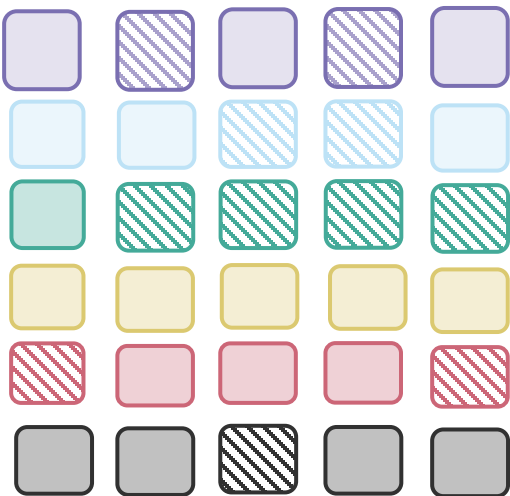
Challenges in Time Series Data Analyses



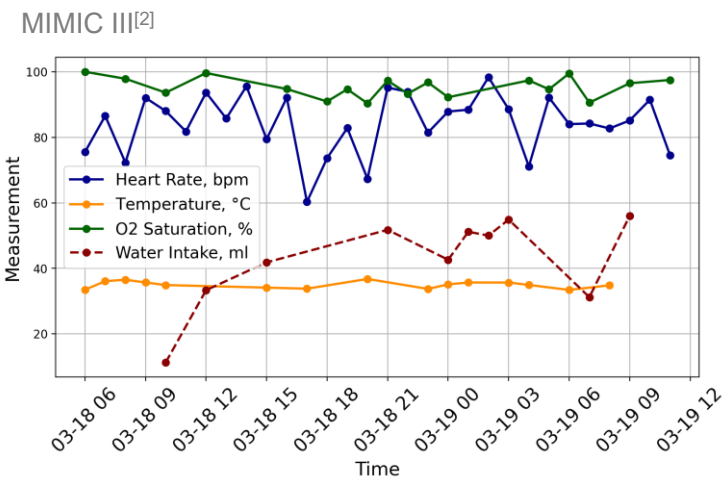
Challenges in Time Series Data Analyses



Heterogeneity

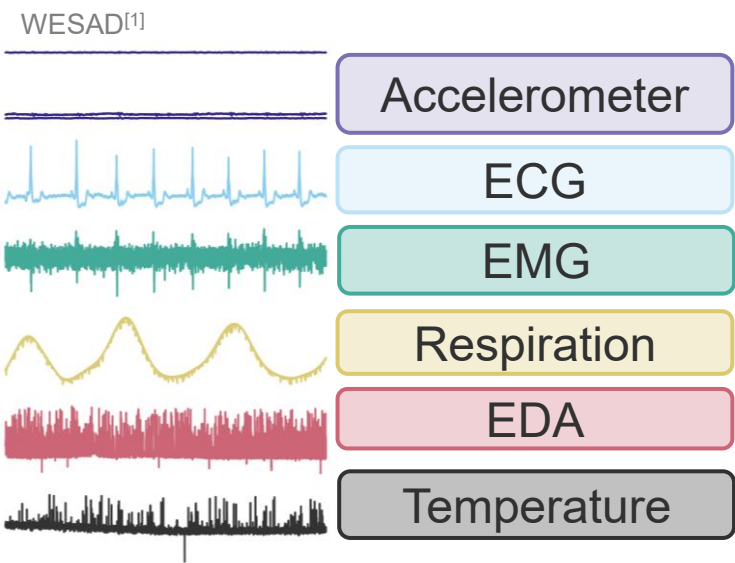


Missingness

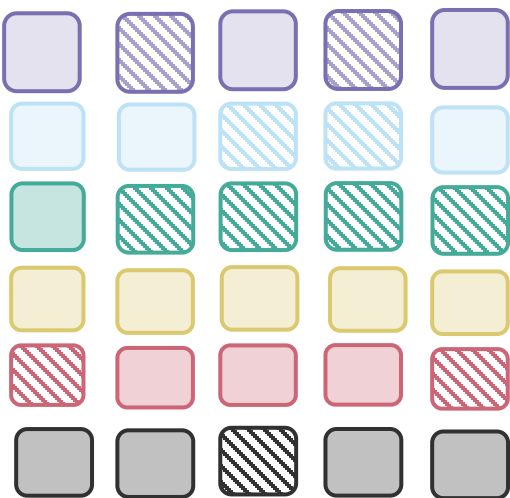


Irregularity

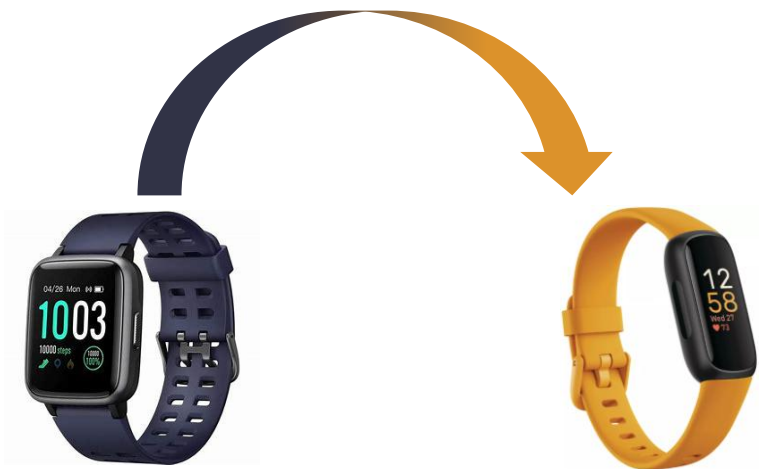
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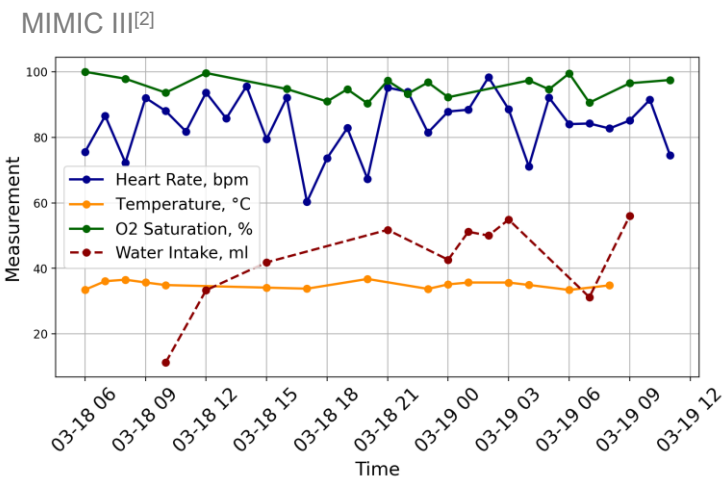
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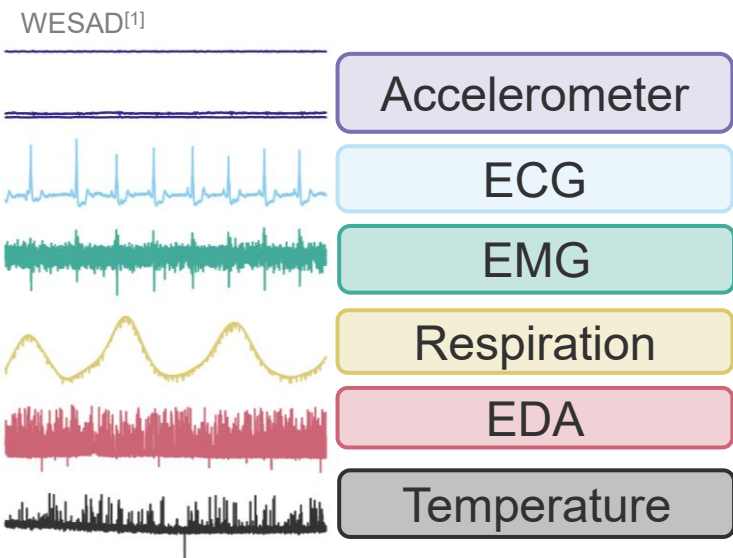


Device Change/Distribution Shift

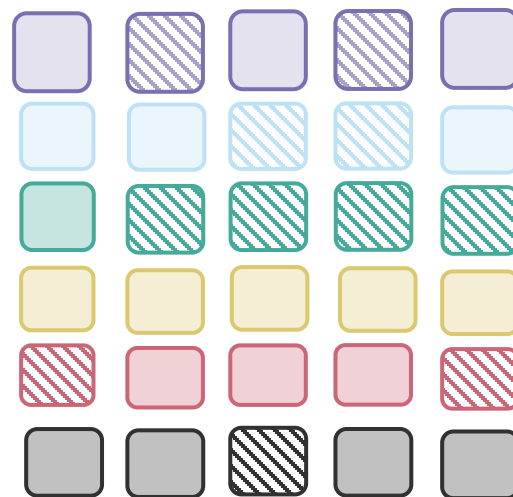


Irregularity

Challenges in Time Series Data Analyses



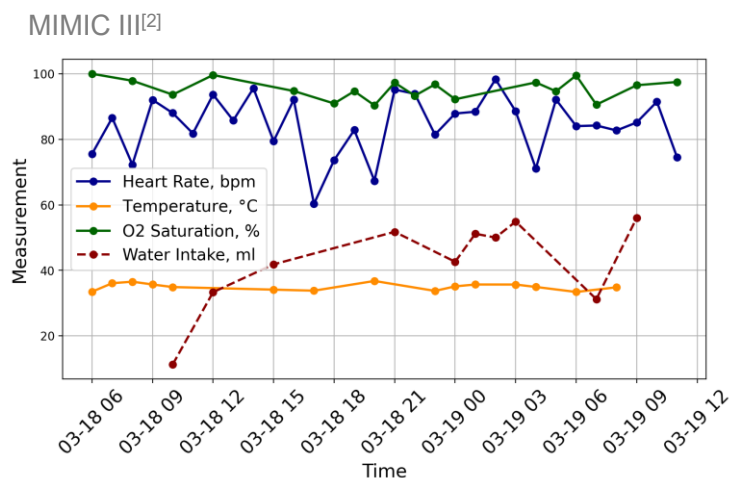
Heterogeneity



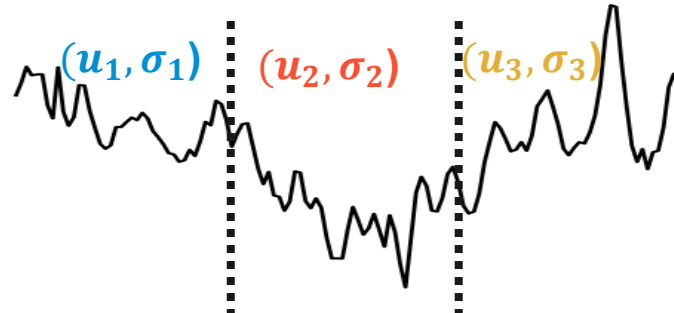
Missingness



Device Change/Distribution Shift

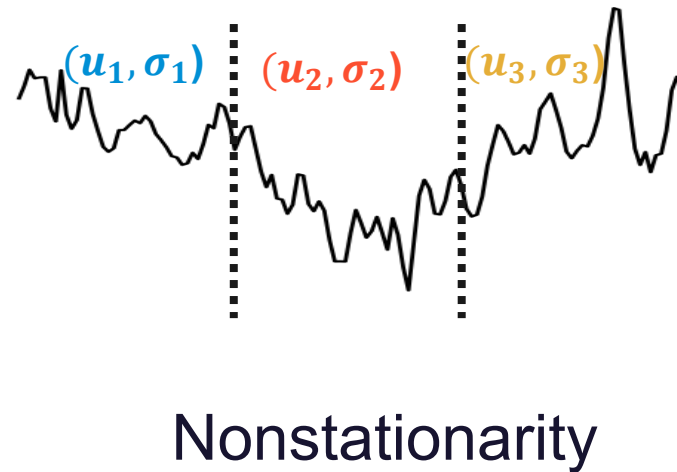
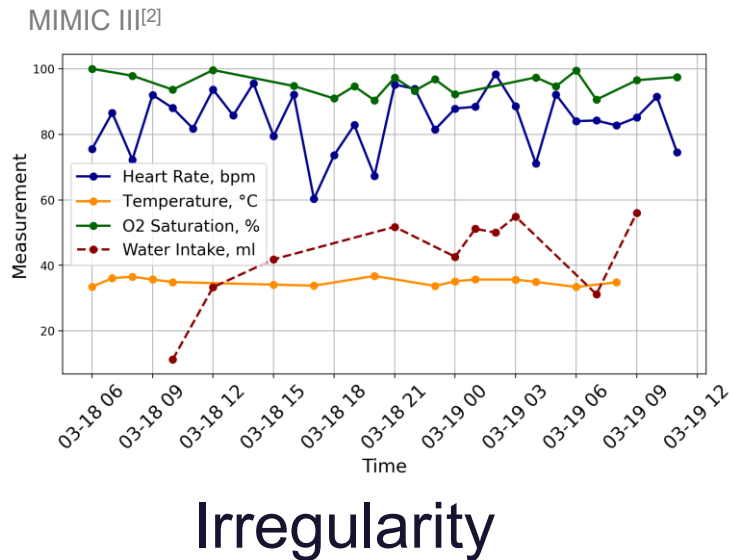
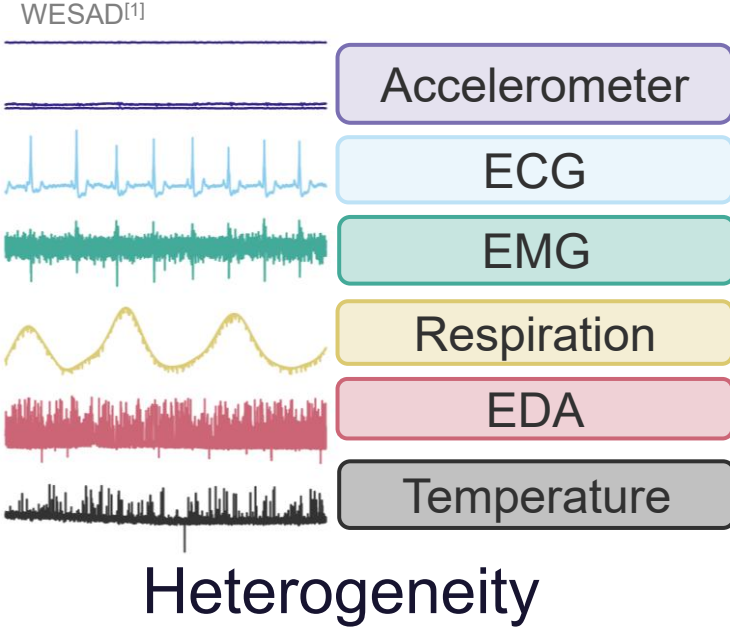


Irregularity

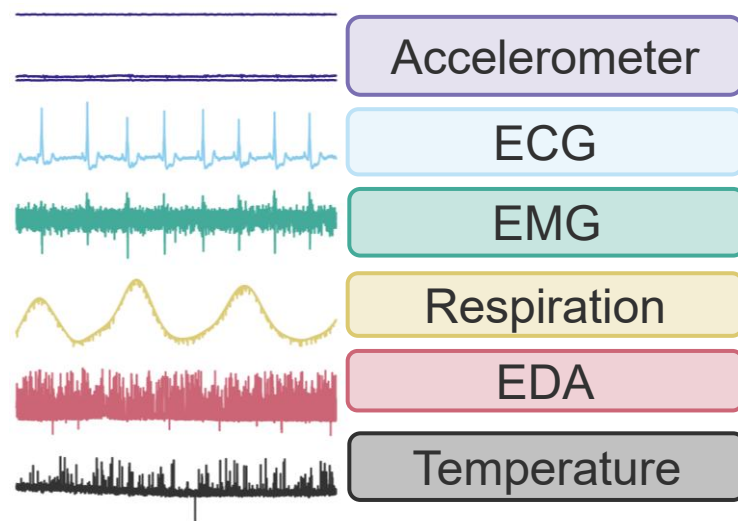


Nonstationarity

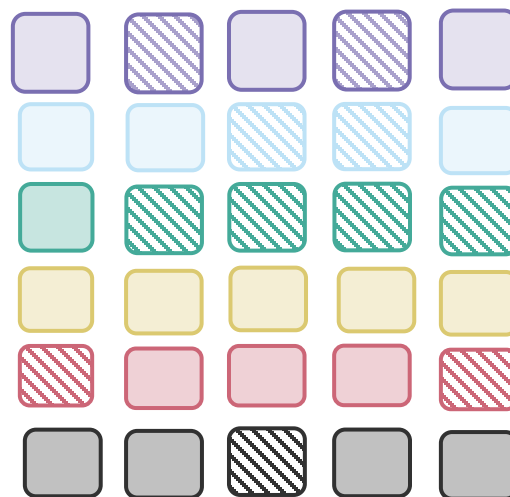
Challenges in Time Series Data Analyses



Agenda



Heterogeneity

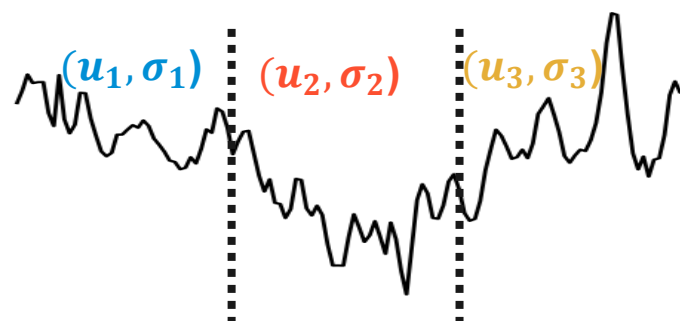


Missingness

I. Robustness to Missing Modalities.



Device Change



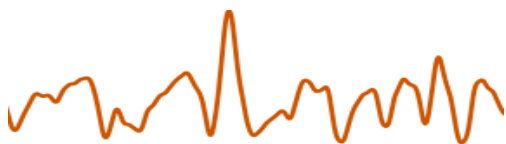
Nonstationarity

II. Robustness to Distribution Shifts.

Traditional Treatment of Time-series



Electrodermal Activity (EDA)



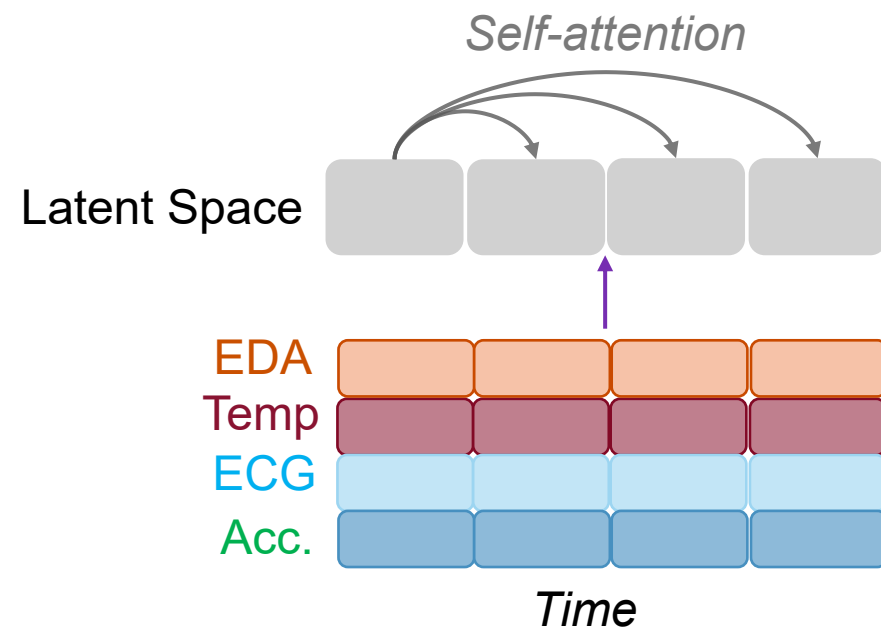
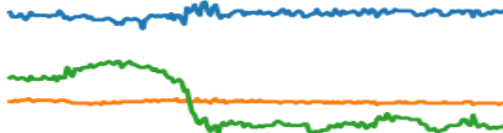
Skin Temperature (Temp)



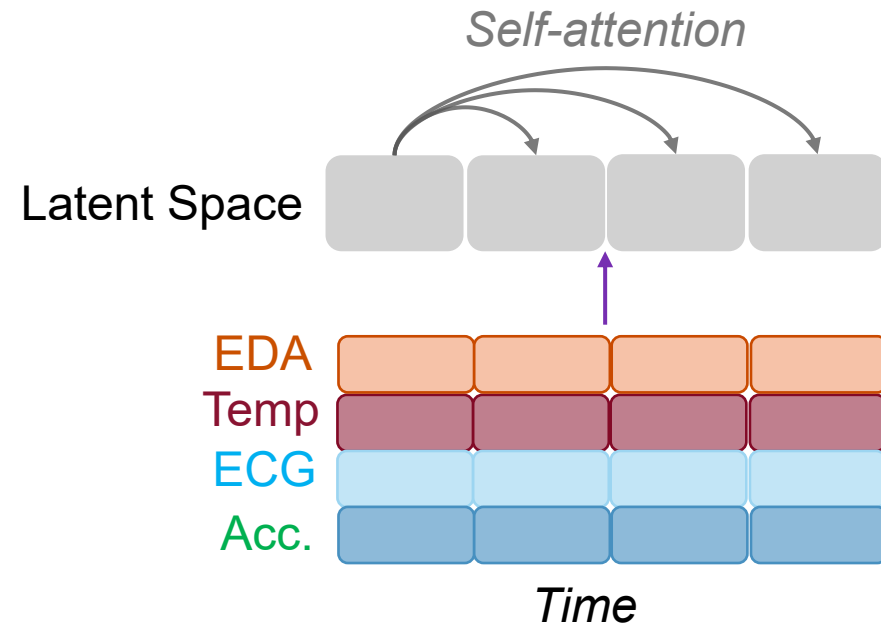
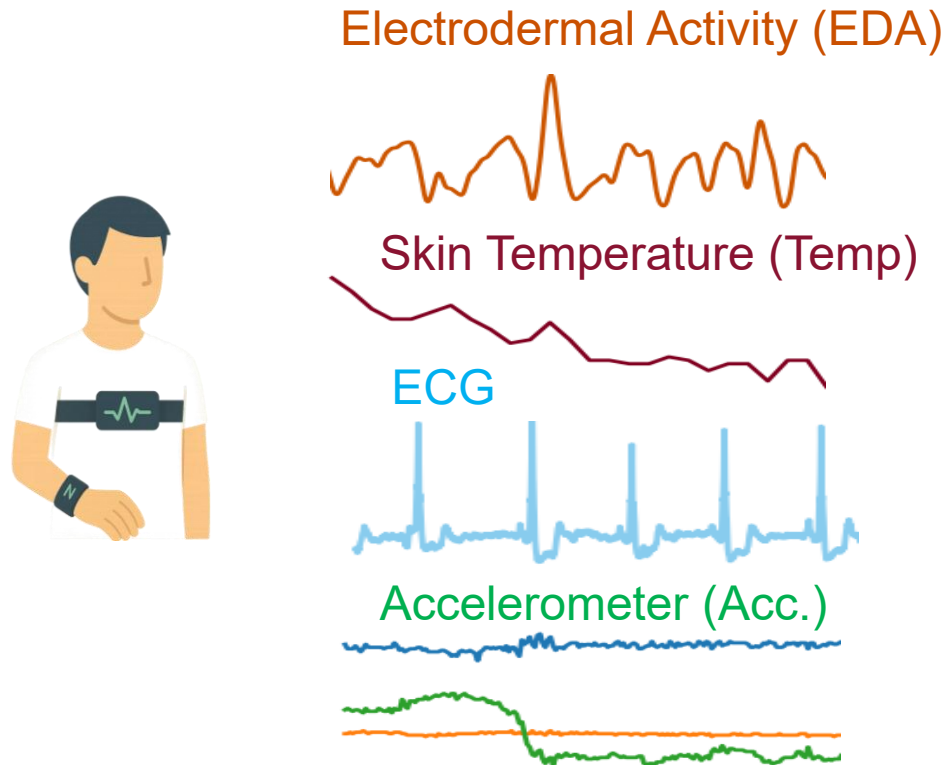
ECG



Accelerometer (Acc.)



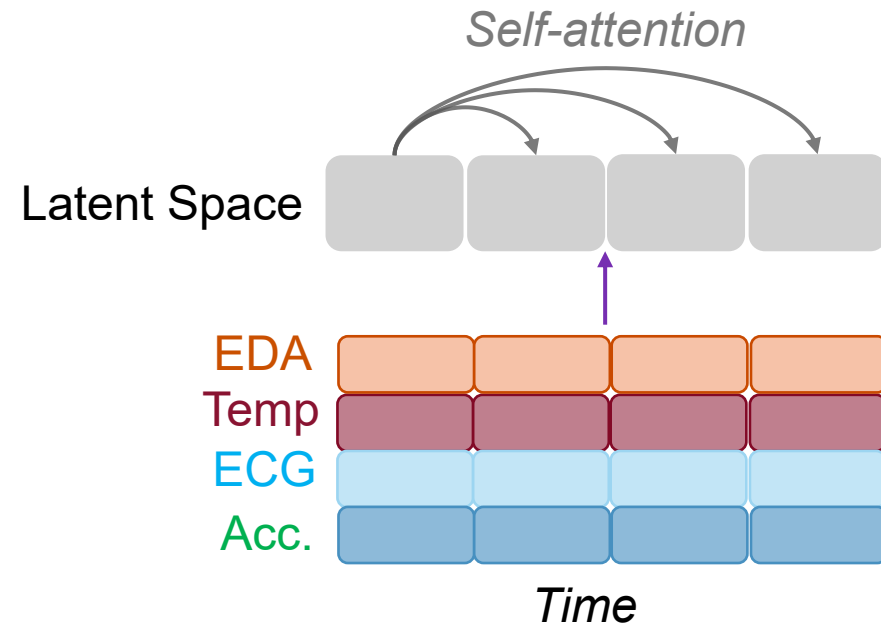
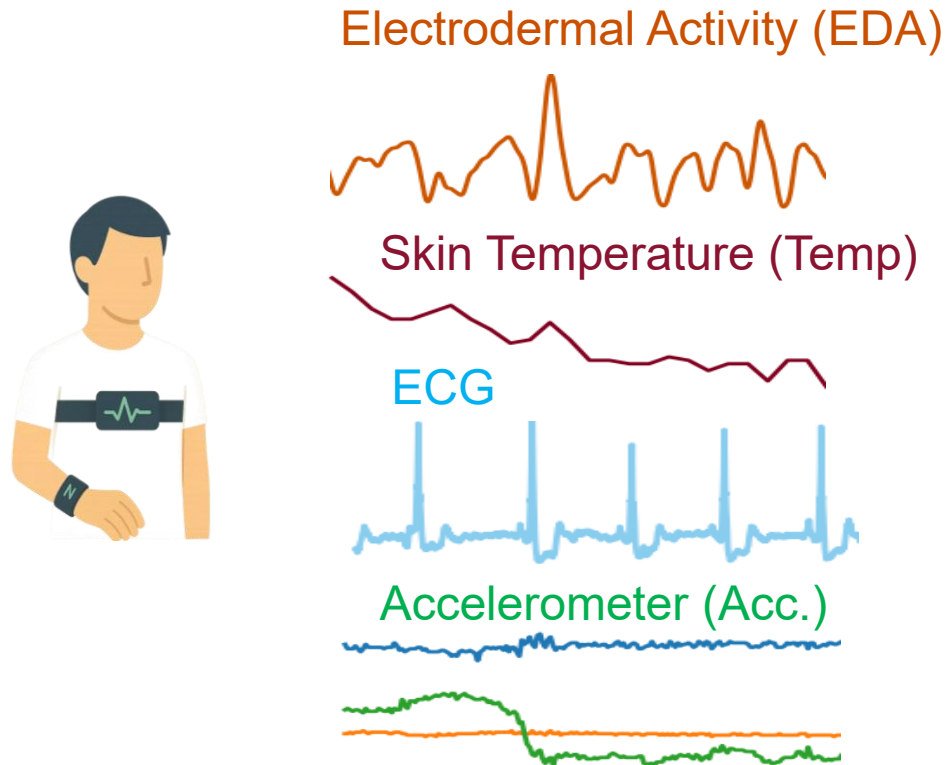
Traditional Treatment of Time-series



While effective in simple tasks :

- Does not model inherent heterogeneity
- Misses modeling inter-modal interactions effectively
- Cannot disentangle representation in case one modality is missing/corrupted.
- Cannot handle different sequence length across modalities.

Traditional Treatment of Time-series



Another approach is sensor fusion, but it is highly application-specific and often heuristic.

For example, in heart rate monitoring from wearables, several proposals have been made in the literature, such as:

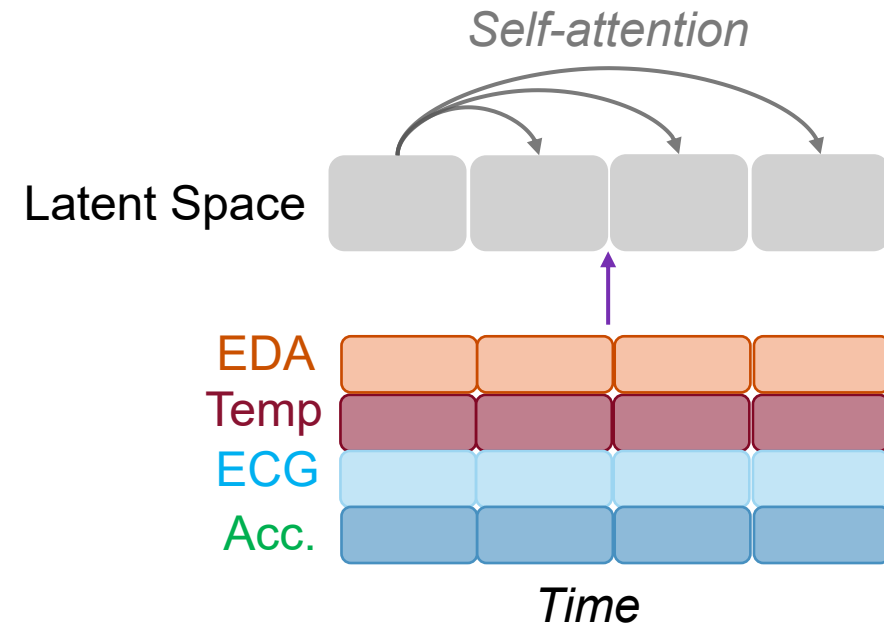
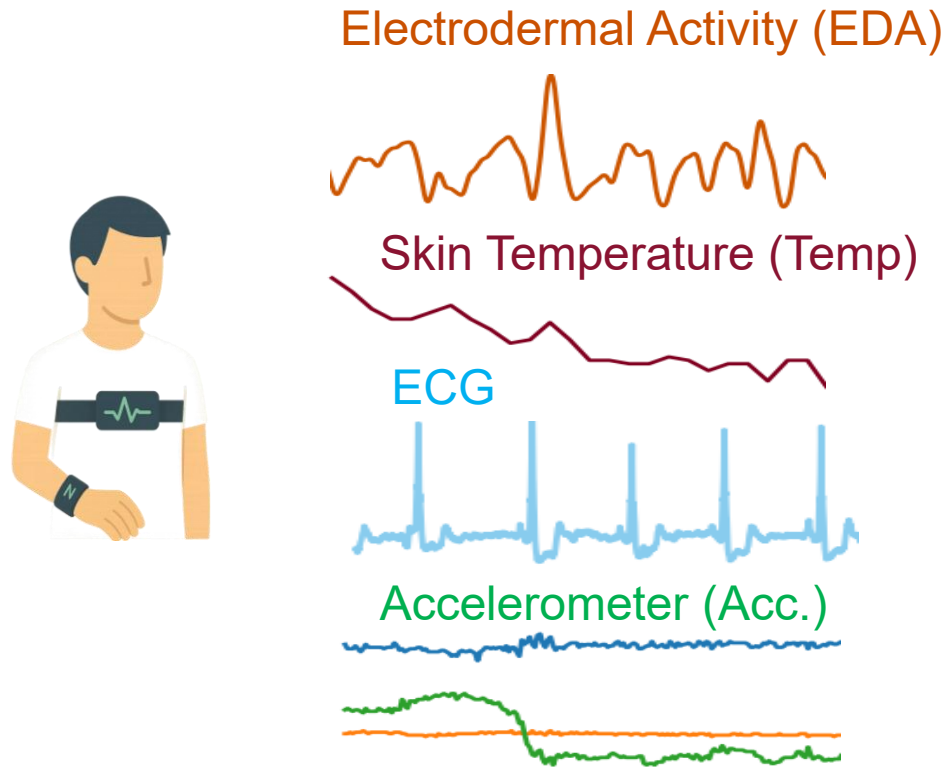
- *early fusion using multi-wavelength photoplethysmography (PPG)¹,*
- *multi-site PPG², and*
- *late fusion with temperature³.*

[1] Meier & Holz (2024). *Effect of wavelength on PPG reliability outdoors* — CISS.

[2] Meier & Holz (2024). *PPG accuracy across body locations and motion* — CISS.

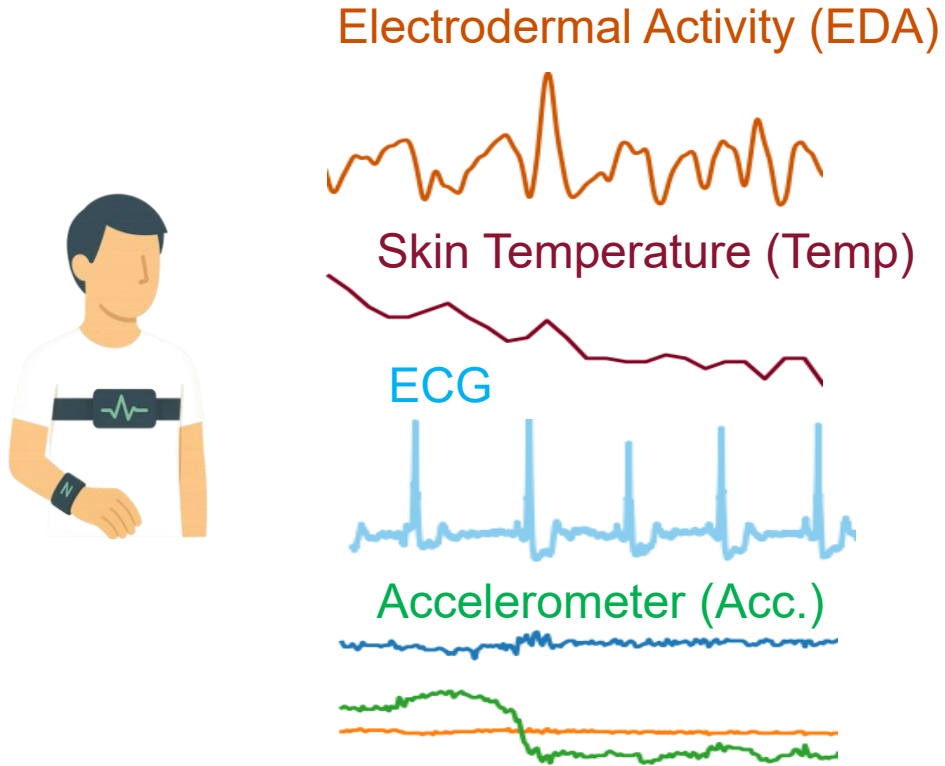
[3] Meier, Demirel & Holz (2024). *WildPPG: Real-world long-duration PPG dataset* — NeurIPS.

Traditional Treatment of Time-series



Need to view as multimodal data
instead of multivariate-time-series!

Traditional Treatment of Time-series

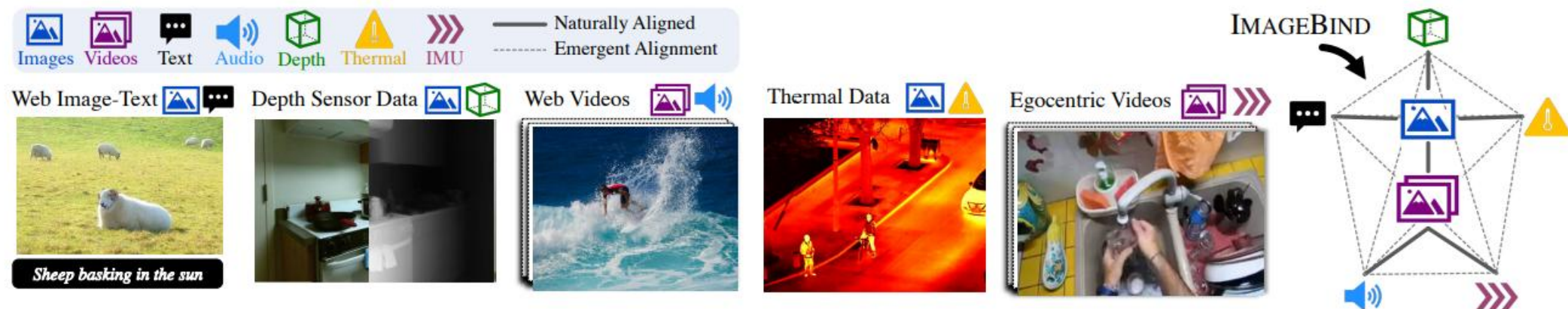


Generally, video, audio, and text representations are considered multimodal. However, even though time-series data are represented in the same numerical form, they can be highly heterogeneous.

Need to view as multimodal data instead of multivariate-time-series!

Common multimodal learning paradigms

I. Binding to an anchor modality^[1]

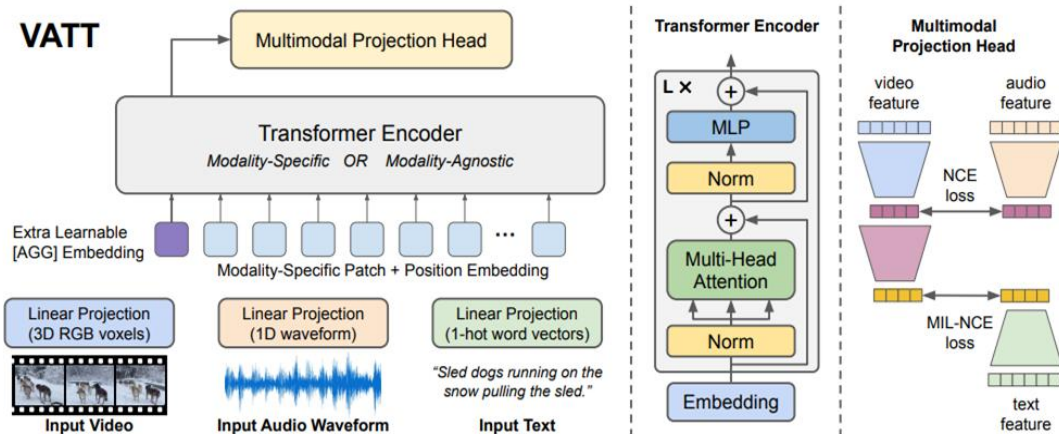


- Need to identify primary modality.
- Primary modality should be always available.

Common multimodal learning paradigms

II. Pairwise similarity matching^[2]

- Structurally contrastive losses work for a pair of inputs (ongoing works on factorized contrastive losses^[3]).
- Assumption of high mutual information (redundancy) among modalities.



[2] VATT, Neurips 2021

[3] Factorized Contrastive Learning, Neurips 2023

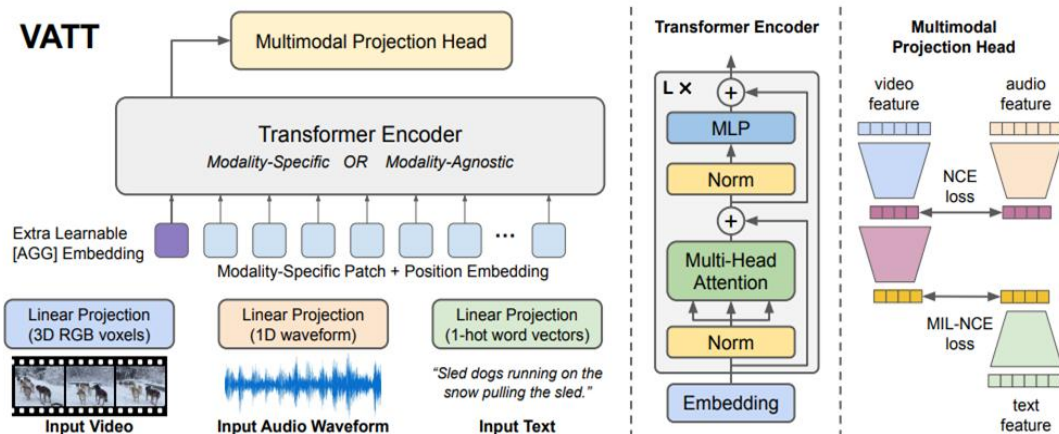
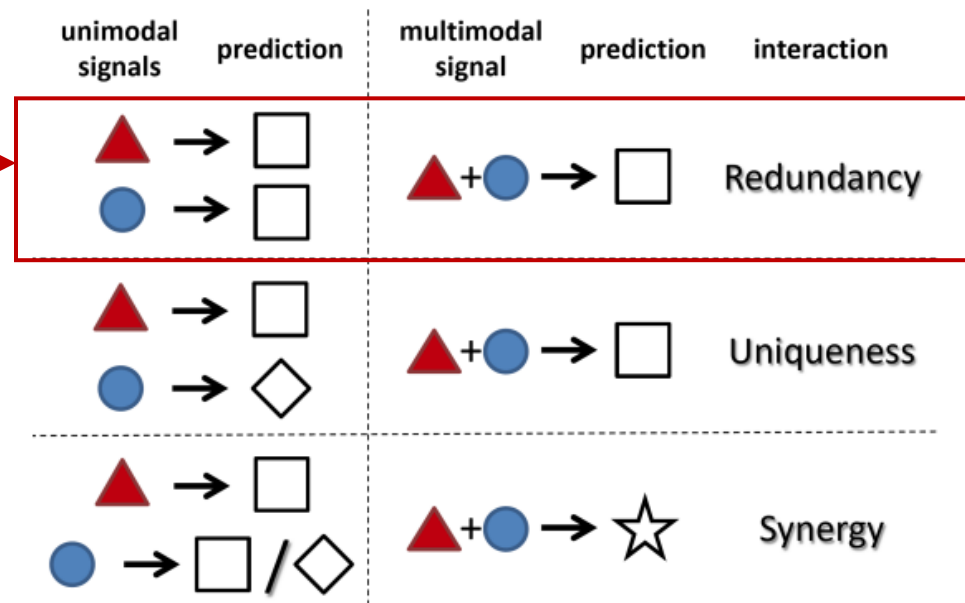
Common multimodal learning paradigms

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[4]



[2] VATT, Neurips 2021

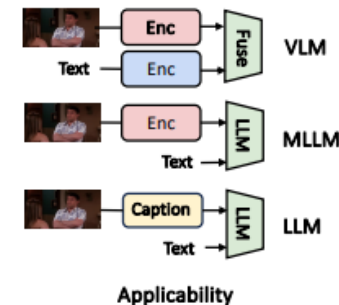
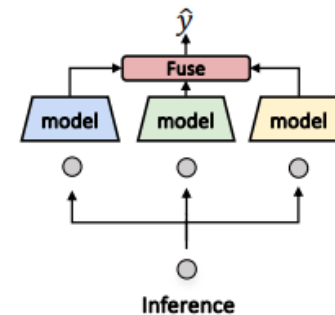
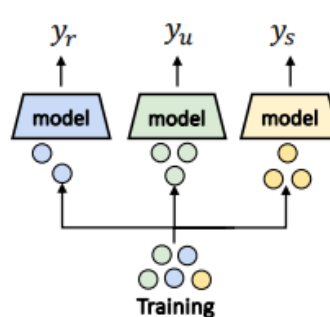
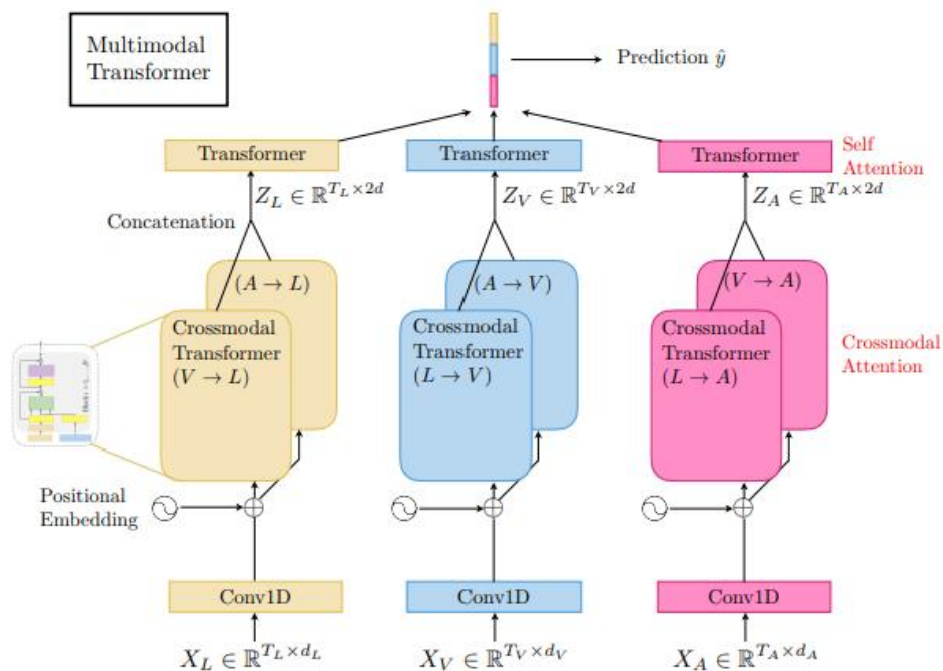
[3] Factorized Contrastive Learning, Neurips 2023

[4] Multimodal Fusion Interactions, ICMI 2023

Common multimodal learning paradigms

III. Explicit cross-modal interaction modeling^[5, 6]

↪ Suitable for small number of modalities to conduct exhaustive pair-wise analysis.



[5] MULT, ACL 2020

[6] MMOE, EMNLP 2024

Considerations for Multimodal Real-world Time-series

→ Need to identify primary modality.

→ Assumption of high mutual information among modalities.

→ Pairwise interaction modeling.

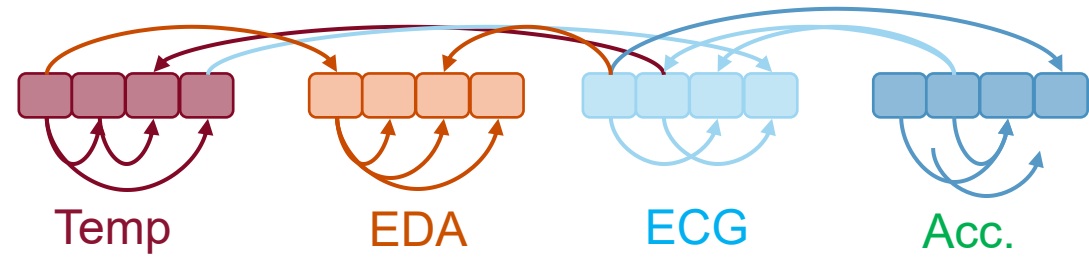
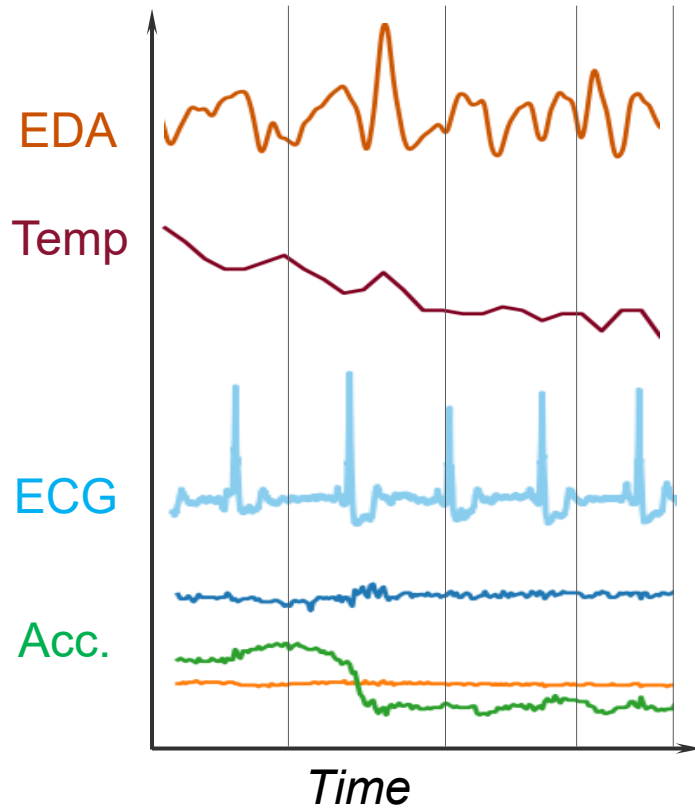
🔒 *Apriori* of primary modality is not always guaranteed.

🔒 Heterogenous modalities.

🔒 Number of modalities can be greater than 10. Combinatorially expansive!

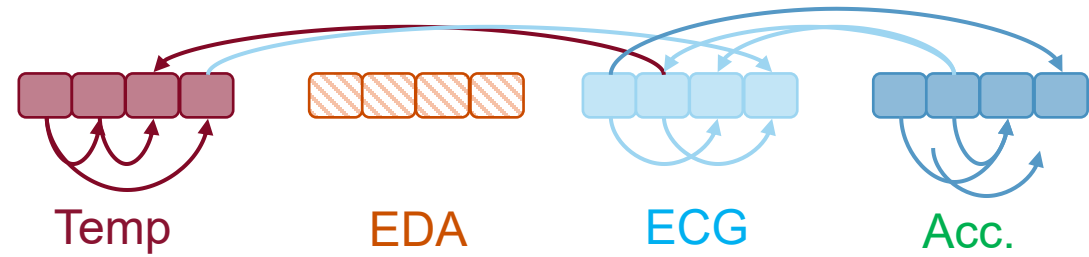
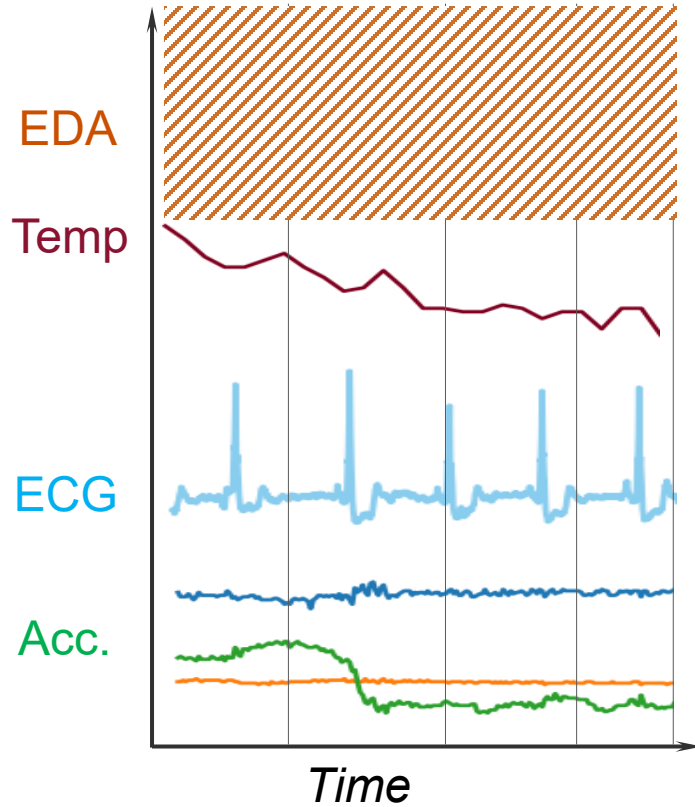
🔒 Random missingness due to sensor malfunction.

Cross-modal-attention for Multimodal Time-series



Cross-attention can allow learning task-relevant modality interaction.

Cross-modal-attention for Multimodal Time-series



Cross-attention can allow learning from arbitrary modality combinations.

But applying Cross-modal-attention through Long Multimodal Time-series increases the computational complexity!

Canonical Self-Attention^[1]

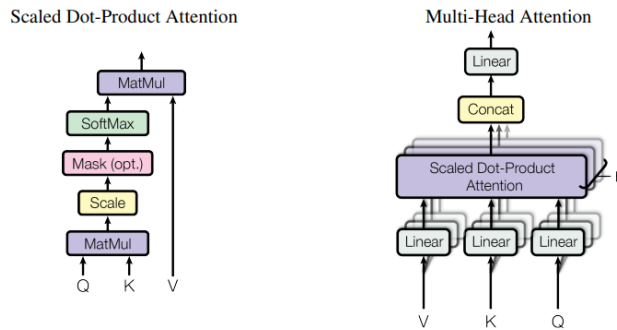


Figure 2: (left) Scaled Dot-Product Attention. (right) Multi-Head Attention consists of several attention layers running in parallel.

$$\mathcal{A}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{Softmax} \left(\frac{\mathbf{Q}\mathbf{K}^{\top}}{\sqrt{d}} \right) \mathbf{V}$$

Point-wise self-attention for a sequence length of L , has a quadratic computational complexity $\rightarrow \mathcal{O}(L^2)$

Consider M modalities each of sequence length L , then the computational complexity increases,

$$\mathcal{O}(M^2 L^2)$$

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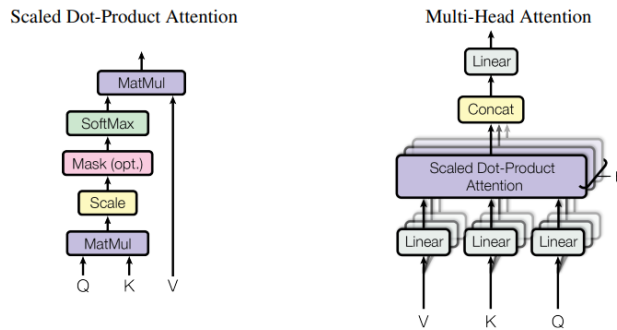


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Consider M modalities each of sequence length L , then the computational complexity increases,

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Sparse
Attention

$$\mathcal{O}(ML \log(ML))$$

Handling long time-series through sparse attention (Overview)

- Point-wise self-attention for a sequence length of L , has a quadratic computational complexity.

$$\mathcal{A}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{Softmax} \left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d}} \right) \mathbf{V}$$

ProbSparse Attention - $\mathcal{A}_s(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{Softmax} \left(\frac{\bar{\mathbf{Q}}\mathbf{K}^\top}{\sqrt{d}} \right) \mathbf{V}$

- Stacking N encoder layers further increases the memory consumption and computational complexity.

$$\hat{s} = s + \text{PE}_{\text{sin}}(s)$$

$$\bar{s} = \mathcal{A}_s(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{Softmax} \left(\frac{\bar{\mathbf{Q}}\mathbf{K}^\top}{\sqrt{d}} \right) \mathbf{V}$$

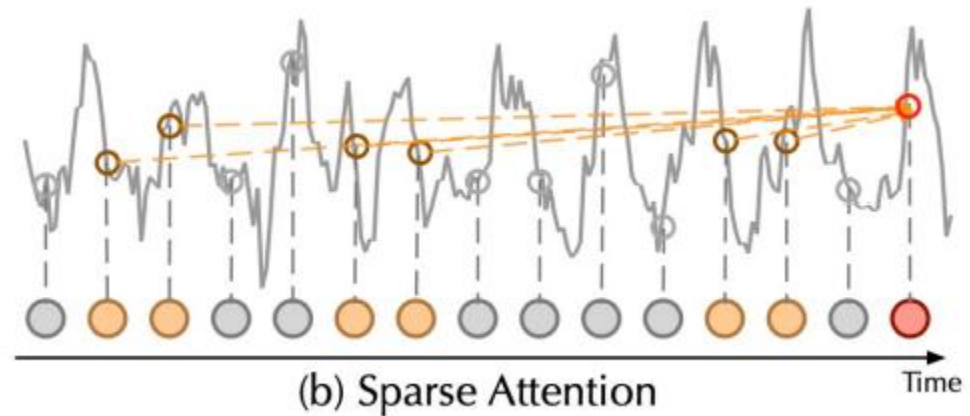
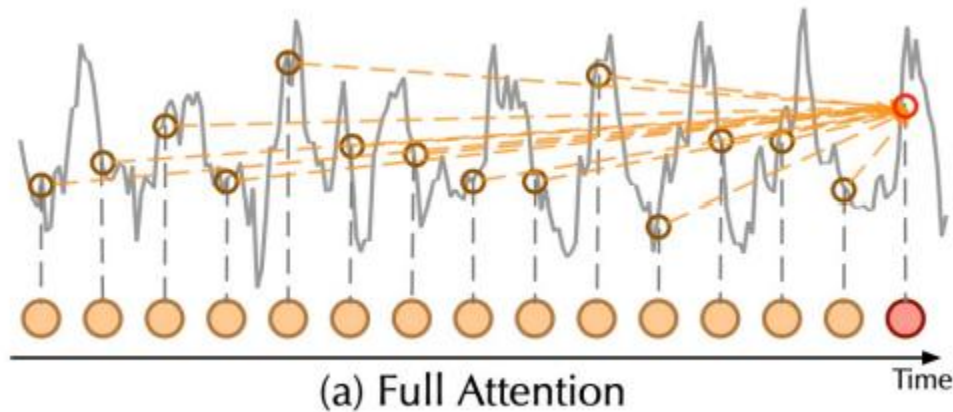
$$\dot{s} = \bar{s} + \hat{s}$$

$$z = \text{distil}(\dot{s}) + \text{maxpool}(\hat{s})$$

Handling long time-series through sparse attention

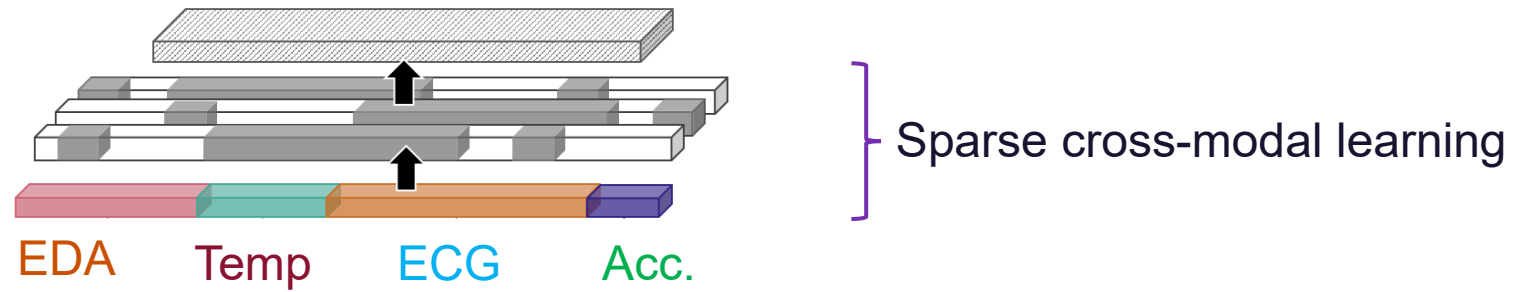
Key Design Motivation :

- Self-attention is long-tailed. So only a few dot-product pairs contribute to the major attention and others can be ignored^[1].
- Using a sparsity metric^[1], we can sample the more informative query-key pairs.
- Instead of L queries, we now use **top- v** where $v = u * \log L$ queries $\rightarrow \mathcal{O}(L \log(L))$



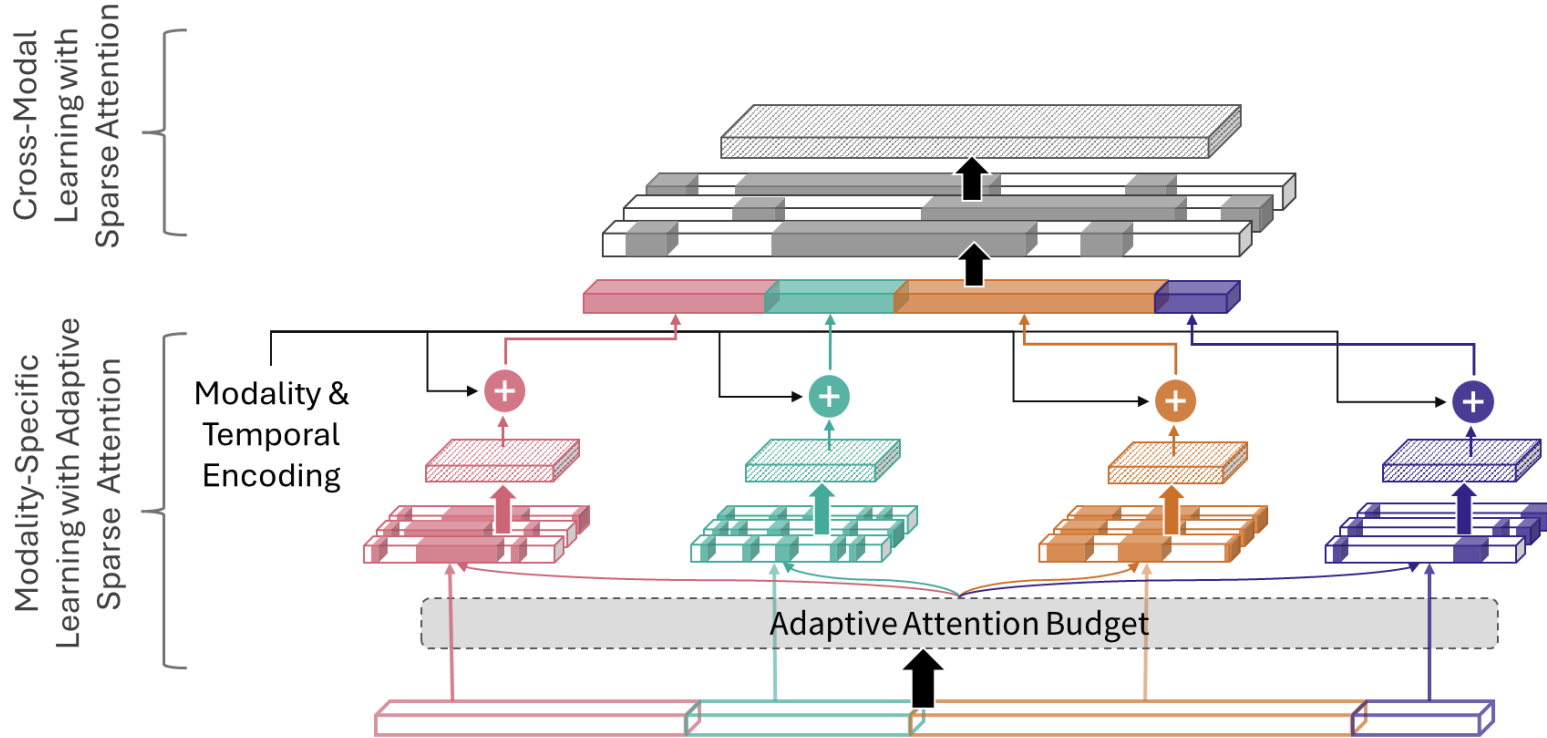
[1] Informer, AAAI 2021

Handling long multimodal time-series through sparse attention



1. Adaptive Attention Budget per modality

Top-v queries in ProbSparse Computation : v is modulated by modality's relevance and availability



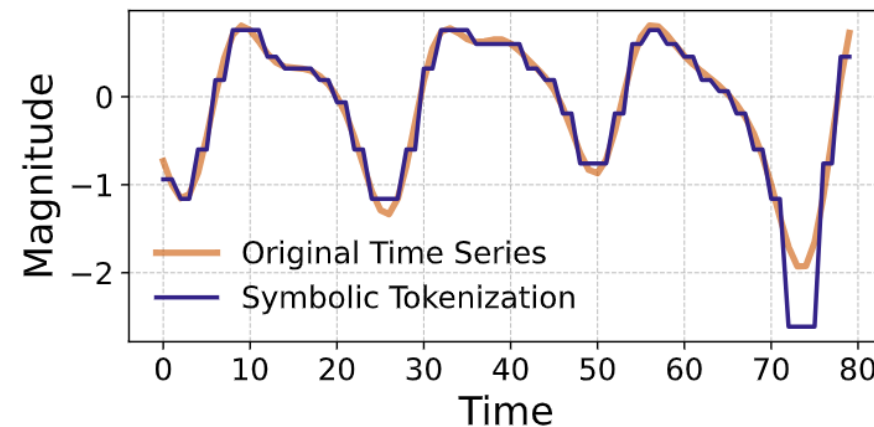
2. Symbolic Tokenization

1. Converts time-series to discrete *symbols*^[1].

- Has some nice properties – guarantees a lower bound Euclidean distance between the symbolic time-series and the original time-series.
- We extend it under some assumptions that this tokenization preserves multimodal relational structure.

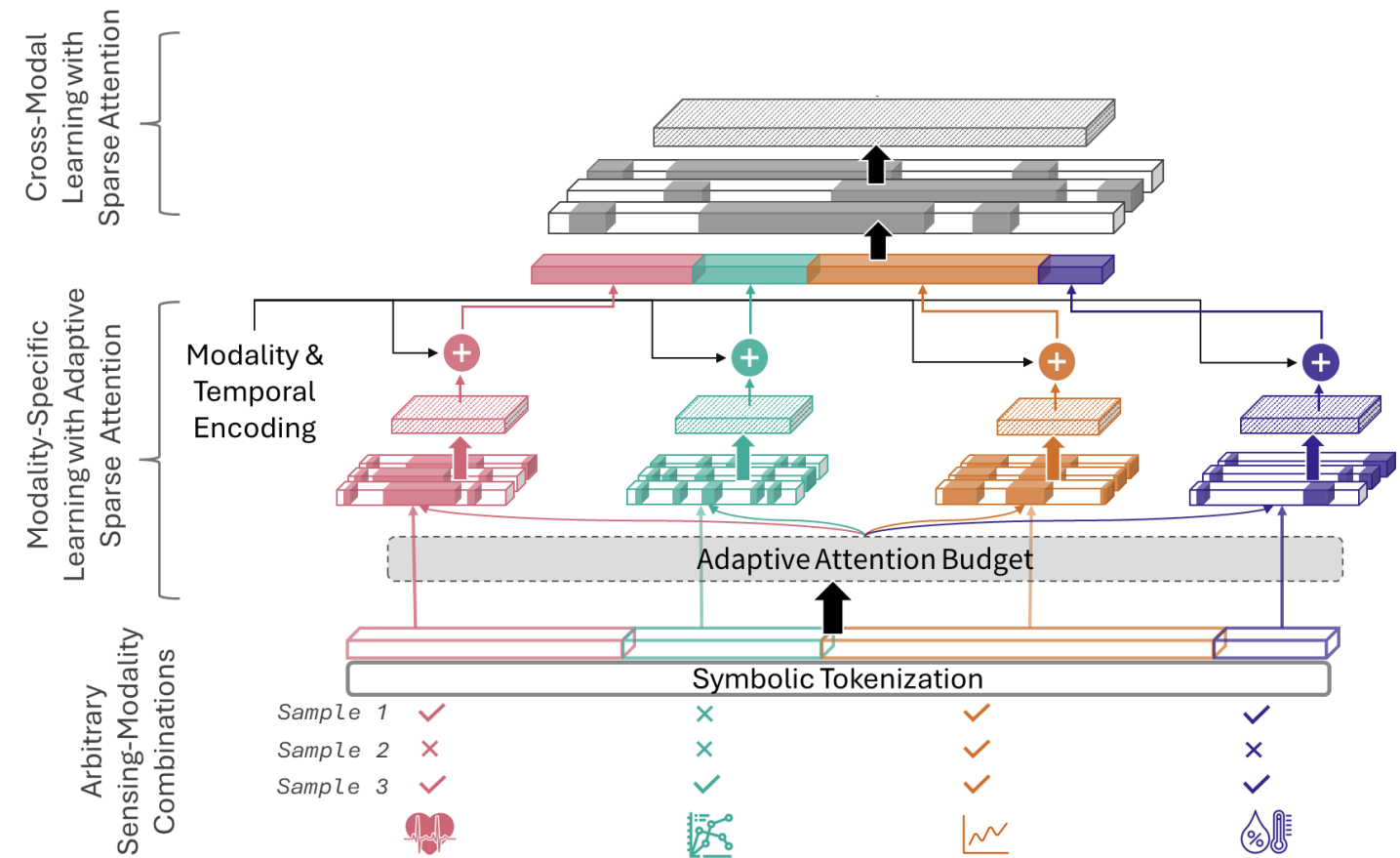
2. We can reserve a symbol for *missing* data naturally.

3. We can compress the signal further reducing the sequence length.



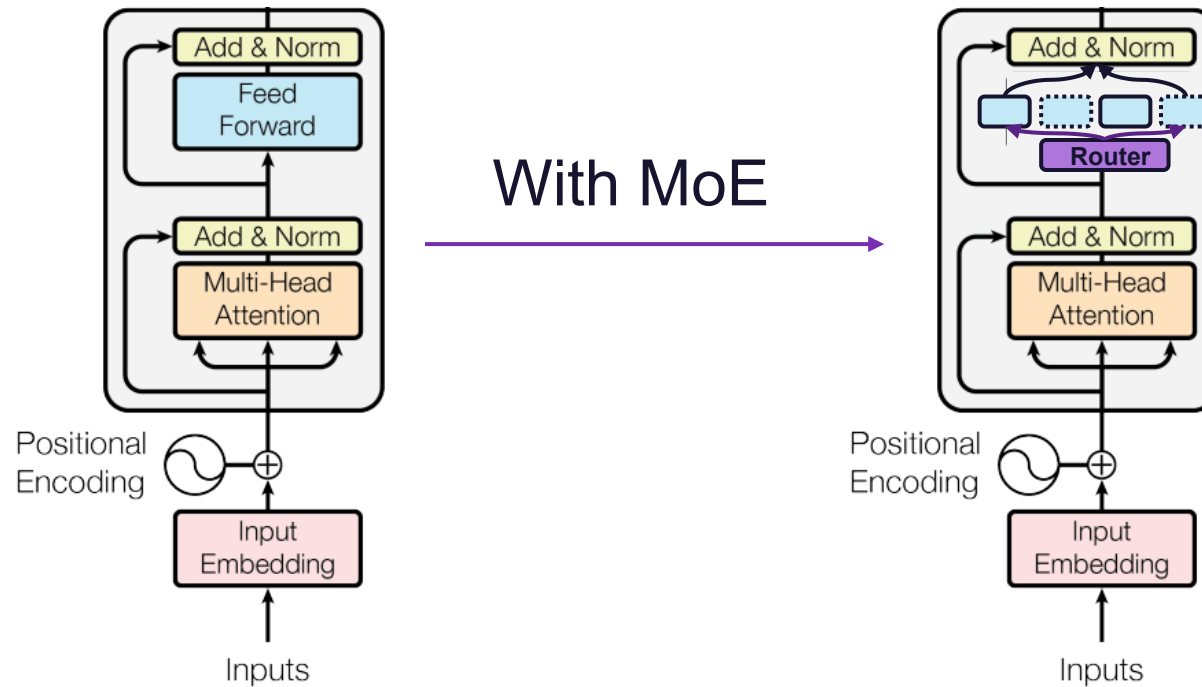
[1] Experiencing SAX: a Novel Symbolic Representation of Time Series, Data Mining and Knowledge Discovery, 2007

2. Symbolic Tokenization



3. Handling Missingness through Mixture-of-Experts

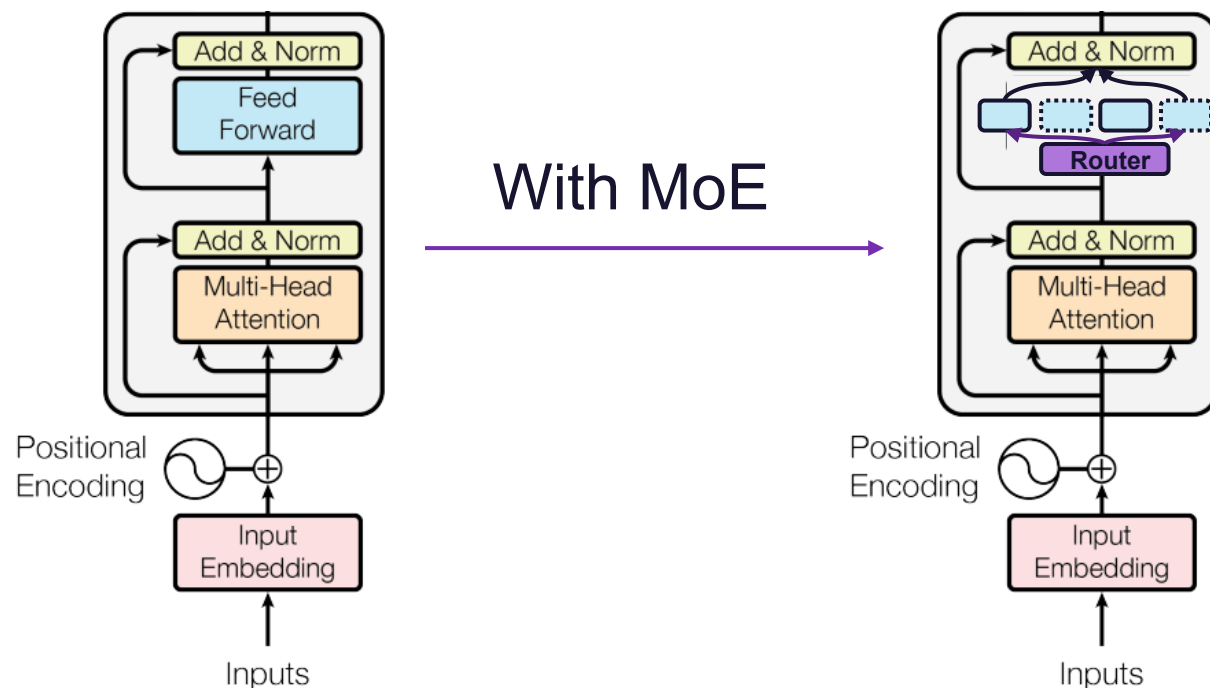
Instead of the fixed Feed-forward layer of the transformer, we can use a mixture of experts(MoE) for implicit modality specialization.



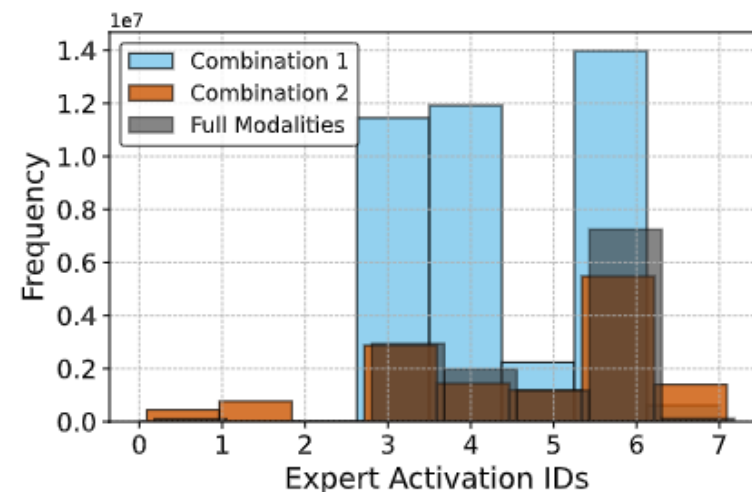
Vanilla Transformer^[1]

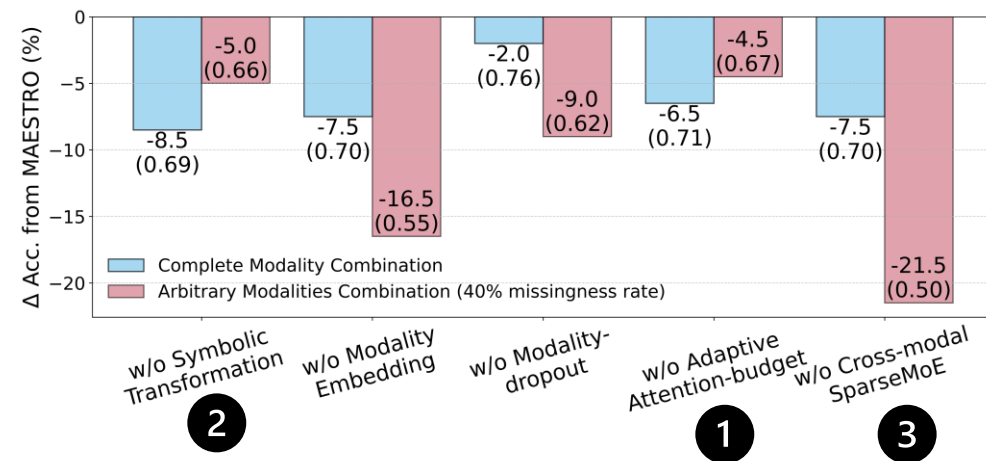
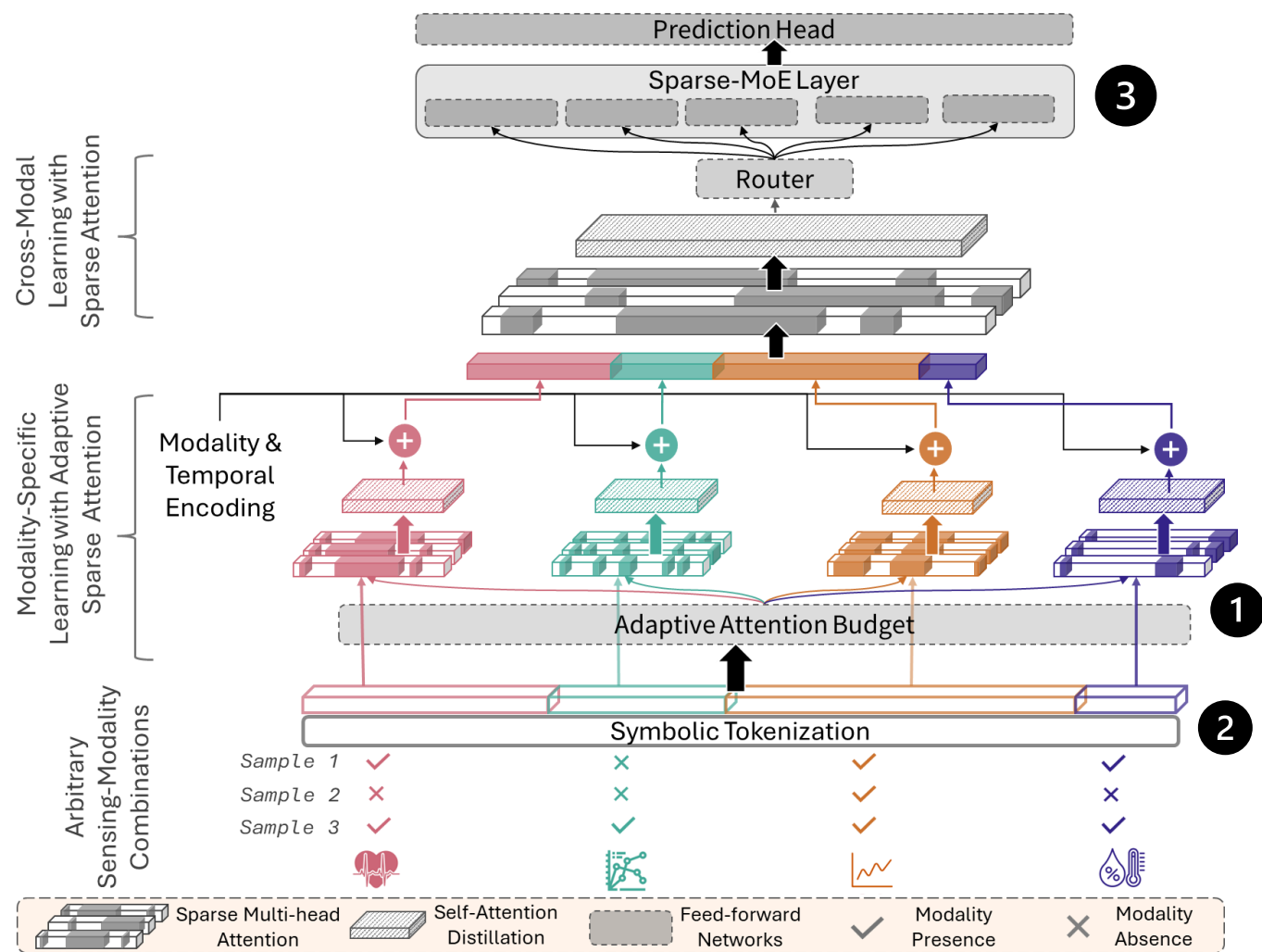
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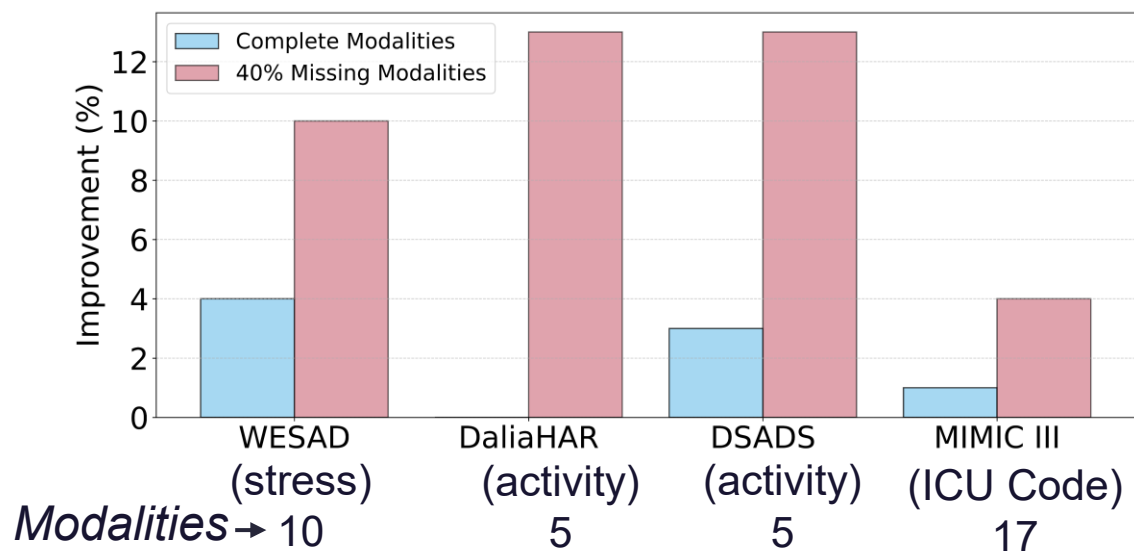




MAESTRO : Adaptive Sparse Attention and Robust Learning for Multimodal Dynamic Time Series, Neurips 2025 (Spotlight 🌟 top 3%)

Key Results : MAESTRO

Average of 4% performance improvement with complete and 8% with arbitrary set of modalities.



Computational Efficiency

Model	Acc. $\uparrow$	MMAC $\downarrow$	GFLOPs $\downarrow$	Params (M)
<i>Multivariate Models</i>				
iTransformer	0.67 ± 0.05	2833	5.73	12.82
Transformer	0.63 ± 0.02	4331	8.66	1.68
<i>Multimodal Models</i>				
FuseMoE	0.47 ± 0.41	6524	13.05	0.67
MULT	0.60 ± 0.42	13324	26.65	3.71
ShaSpec	0.62 ± 0.51	4556	9.11	216
MAESTRO	0.77 ± 0.04	3066	6.13	1.39
– Full-Attn (Per-Modal)	0.80 ± 0.03	3769	7.54	1.40
– Full-Attn (Cross-Modal)	0.77 ± 0.07	3496	6.99	1.39
– All Full-Attention	0.75 ± 0.05	4205	8.42	1.39
– All Full-Attention (no MoE)	0.78 ± 0.04	4392	8.78	1.39



Need for Generalization in Sensing Applications

Different Devices^[1, 2]



Subpopulation Shift^[1, 2]

Different Age-groups

20-40

40-60

60-80

80-99

Human Activity Recognition (HAR)
using
Inertial Measurement Units' recordings.



Sleep-Stage Classification (SSC) using
Electroencephalogram (EEG) Recordings.



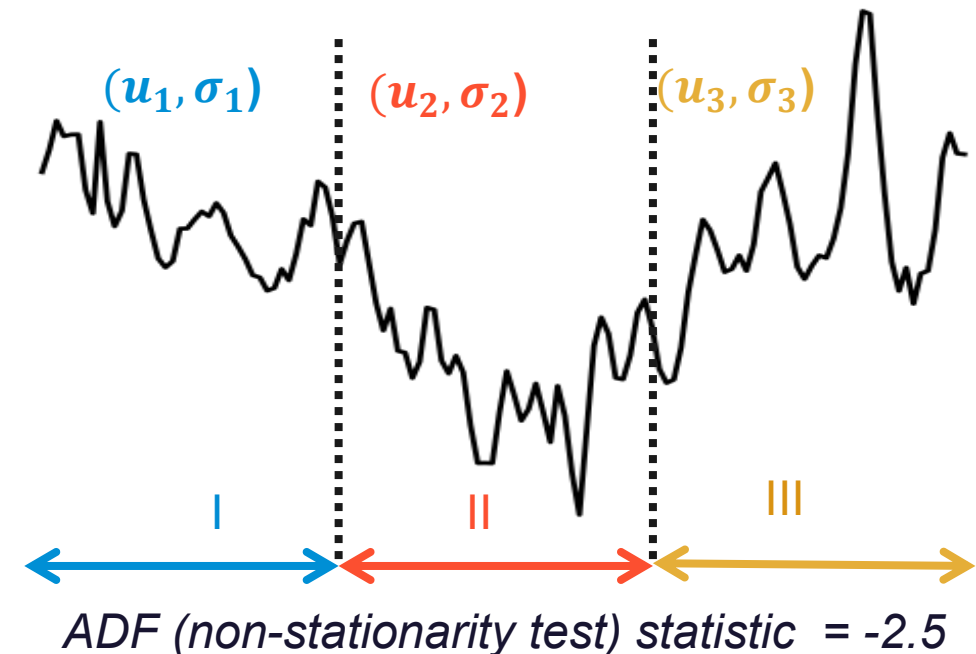
Need for Generalization in Sensing Applications

Not only is there a domain-gap^[1]

	HAR				
	UCIHAR	WISDM	HHAR	SSC	MFD
Same Domain (Target-only)	100.00	98.02	98.55	72.09	99.39
Cross-Domain (Source-only)	65.94	48.60	63.07	51.67	72.51
Gap (δ)	37.32	49.44	33.86	18.38	26.88

But also intra-domain nonstationarity^[2, 3].

- varying statistical and spectral properties over time.



[1] ADATIME: A Benchmarking Suite for Domain Adaptation on Time Series Data

[2] Out-of-Distribution Representation Learning for Time-Series Classification

[3] AdaRNN: Adaptive Learning and Forecasting of Time Series

Considerations for Real-world Time-series

- ⌘ Applying standard Domain Generalization (DG) algorithms to time-series is ineffective^[1, 2].
 - ⌘ Time-Series Domain Adaption techniques^[3].
 - Need access to some target domain data for alignment.
- ❑ Need for time-series classification method that
- ✓ Learns generalizable representations.
 - ✓ Doesn't access target domain data.
 - ✓ Doesn't require explicit domain labels.



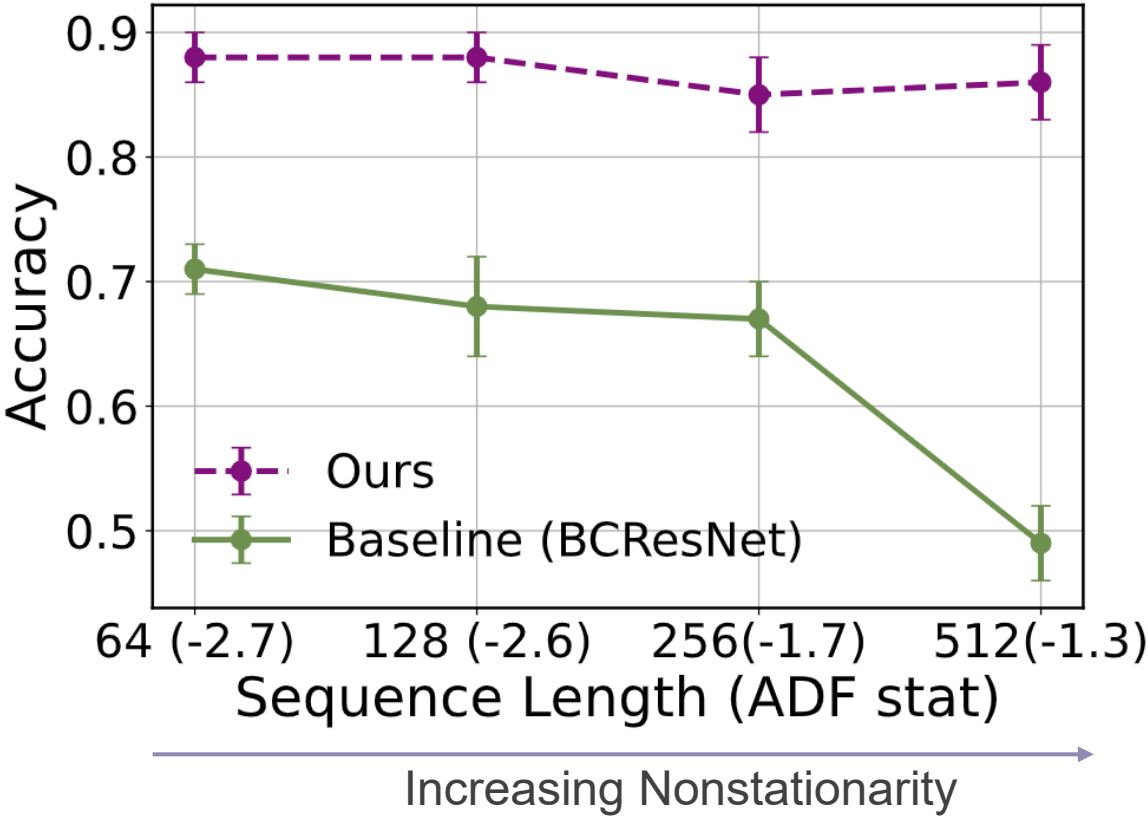
[1] GLOBEM Dataset: Multi-Year Datasets for Longitudinal Human Behavior Modeling Generalization

[2] WOODS: Benchmarks for Out-of-Distribution Generalization in Time Series

[3] Source-Free Domain Adaptation with Temporal Imputation for Time Series Data

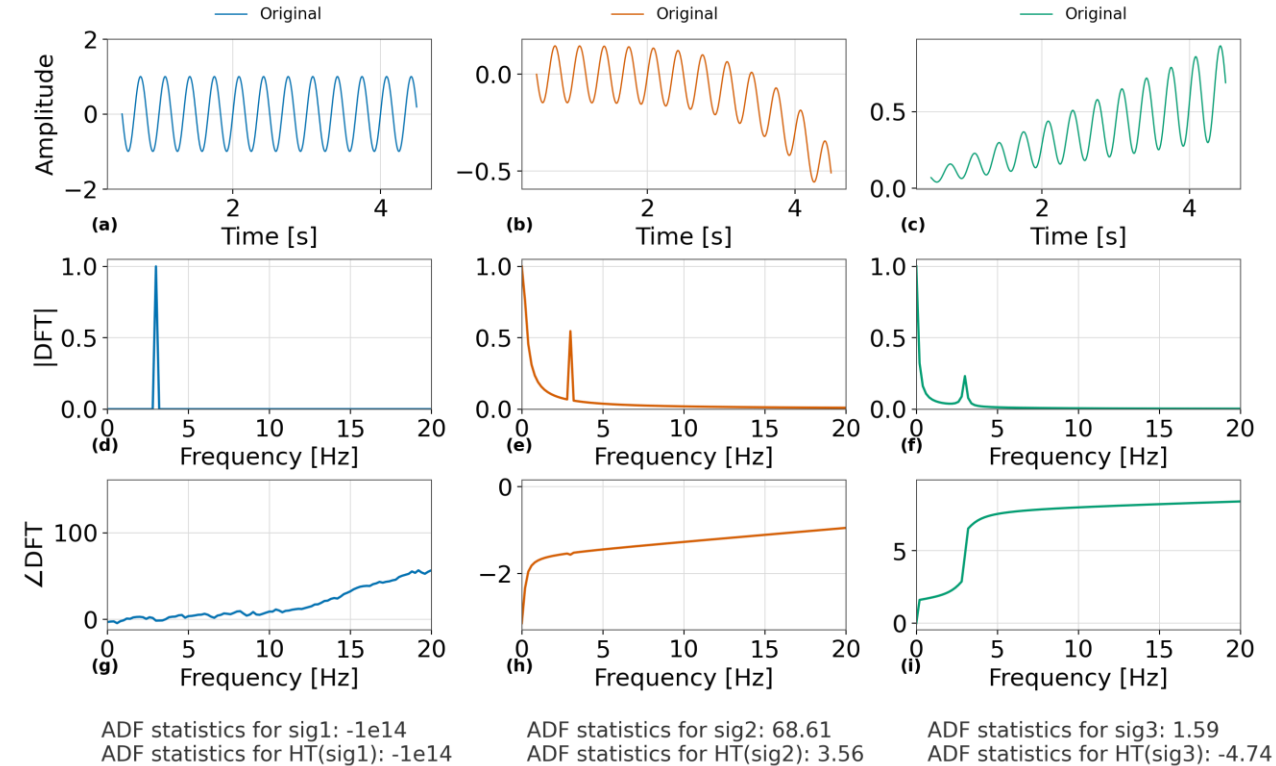
Nonstationarity, Generalization and Phase

1. Nonstationarity impacts generalization.



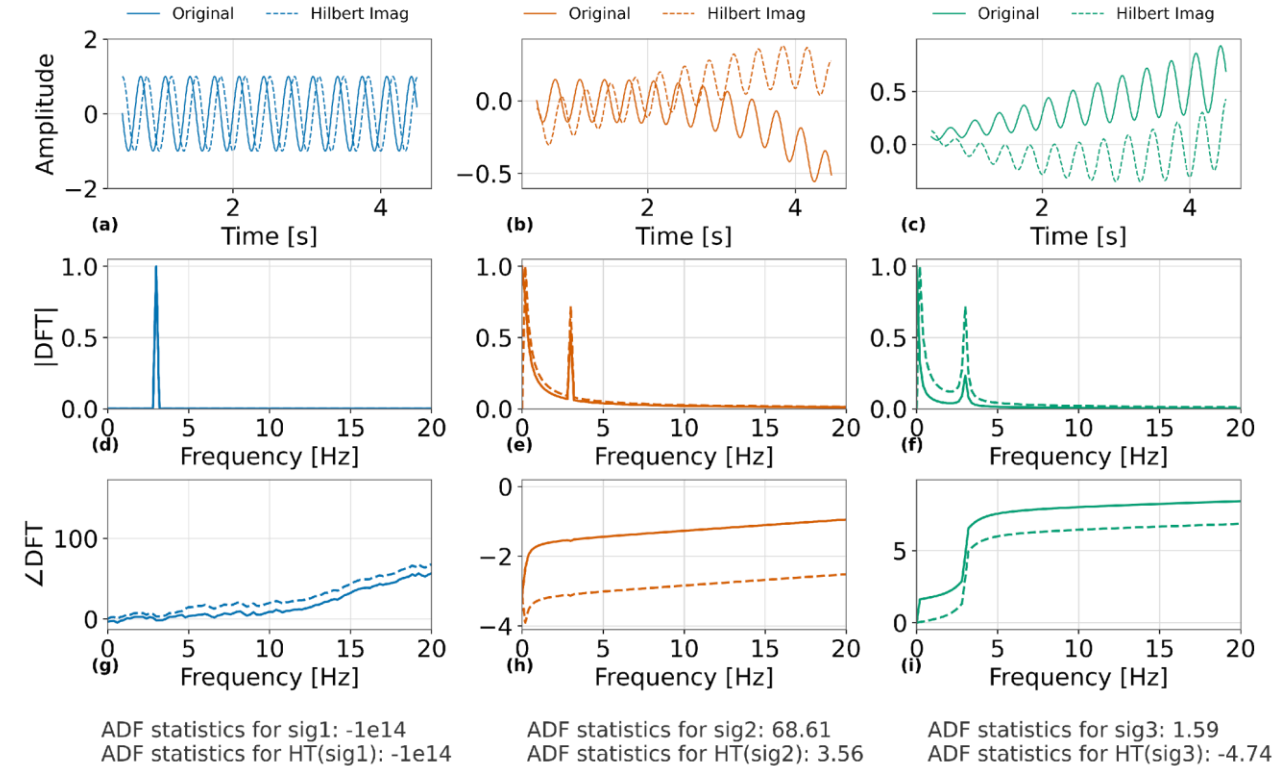
Nonstationarity, Generalization and Phase

1. Nonstationarity impacts generalization.
2. Phase information of a signal encodes nonstationarity.

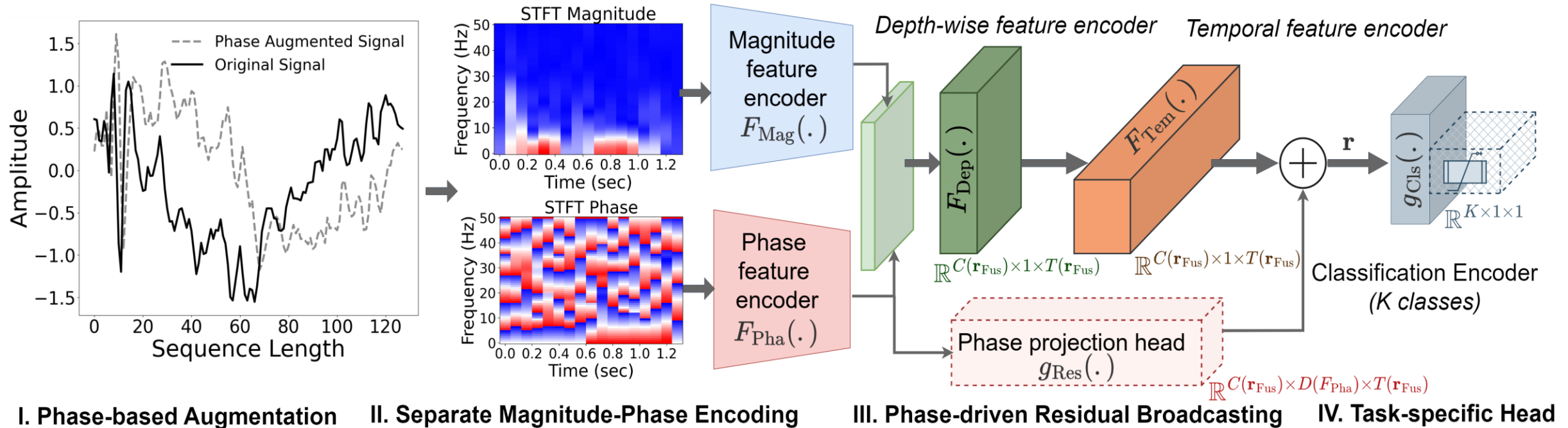


Nonstationarity, Generalization and Phase

1. Nonstationarity impacts generalization.
2. Phase information of a signal encodes nonstationarity.
3. Changing Phase (carefully) can diversify nonstationarity.
 - which can aid in generalization.



Phase-driven Generalizable Representation for Time-series

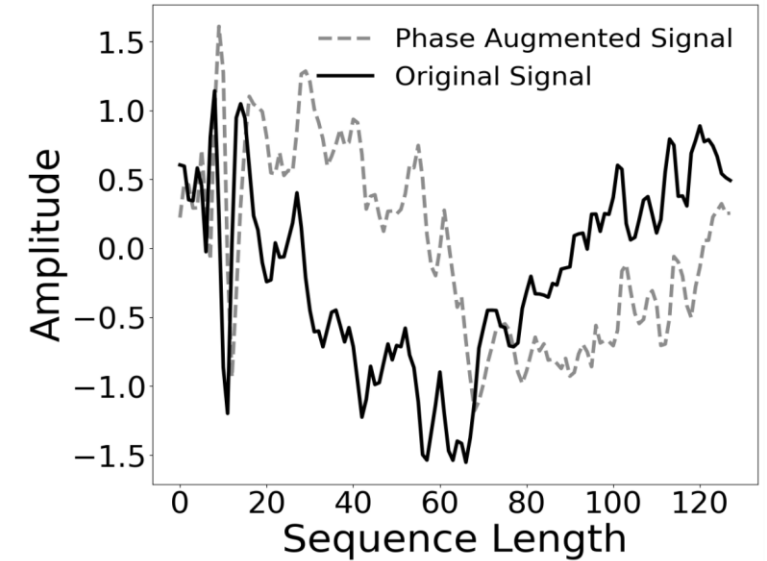


PhASER leverages phase augmentation, separate magnitude-phase encoding, and phase-residual broadcasting for distribution-invariant learning.

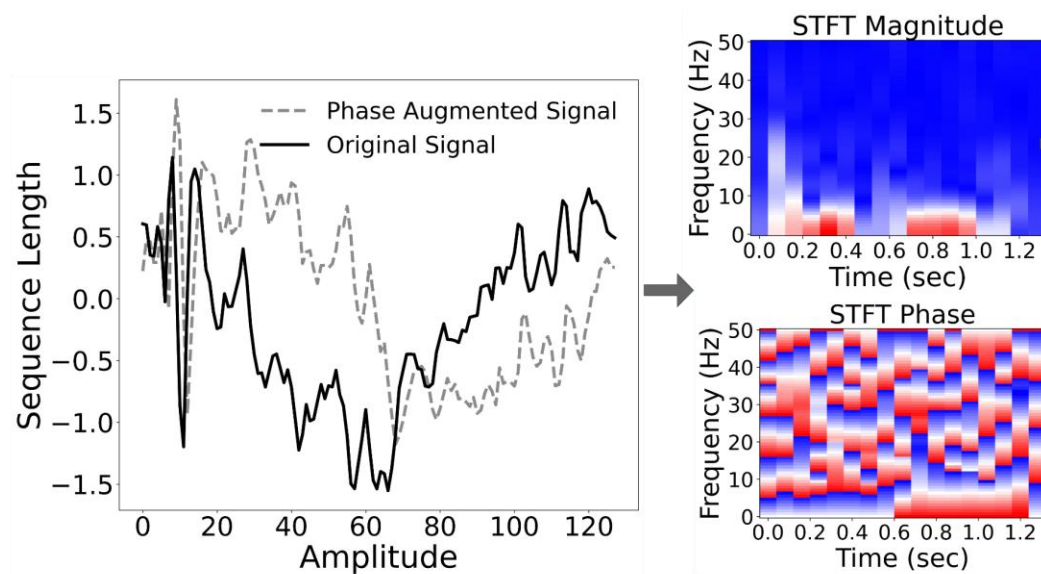
1. Hilbert-transform based augmentation

Interesting properties of Hilbert Transform for us

- Capable of handling nonstationary signals.
- Retains magnitude response.
- In-sample augmentation (unlike mixup and other strategies)
 - No need to incorporate any application-specific characteristics to achieve this.
 - Simple.

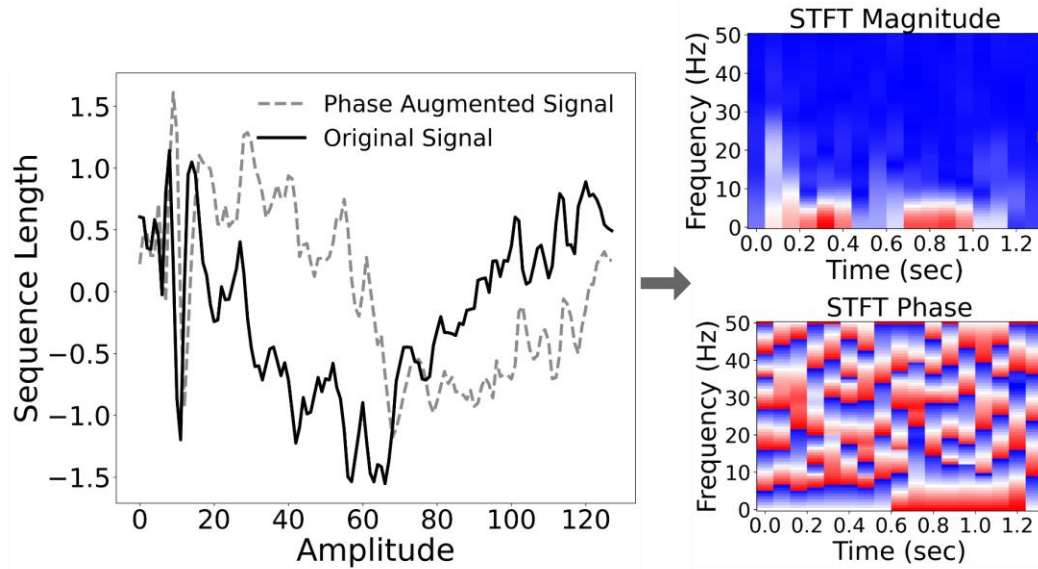


2. Separate Magnitude and Phase Encoding

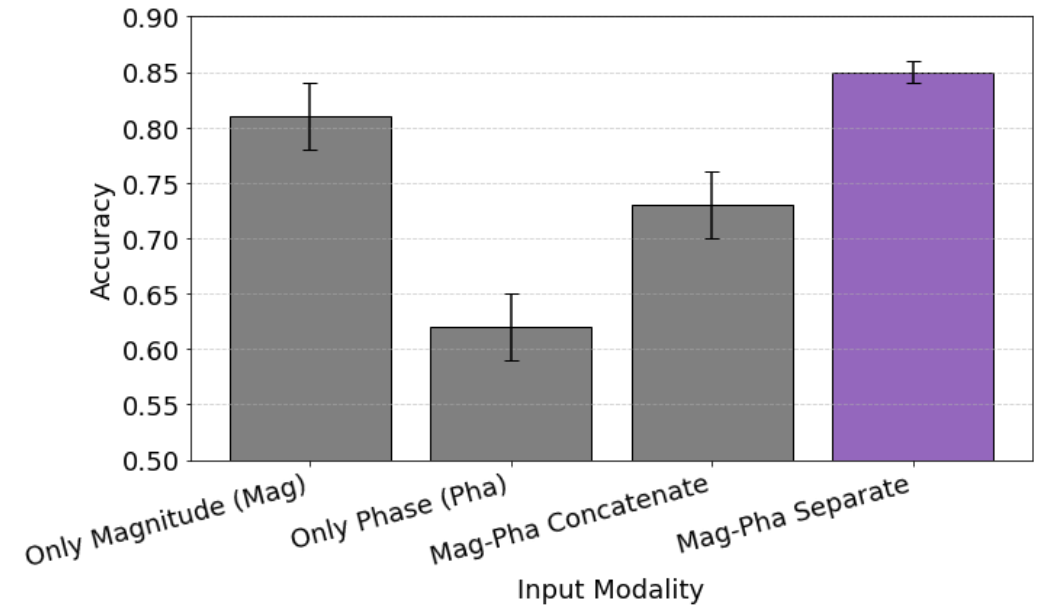


I. Phase-based Augmentation II. Separate Magnitude-Phase Encoding

2. Separate Magnitude and Phase Encoding



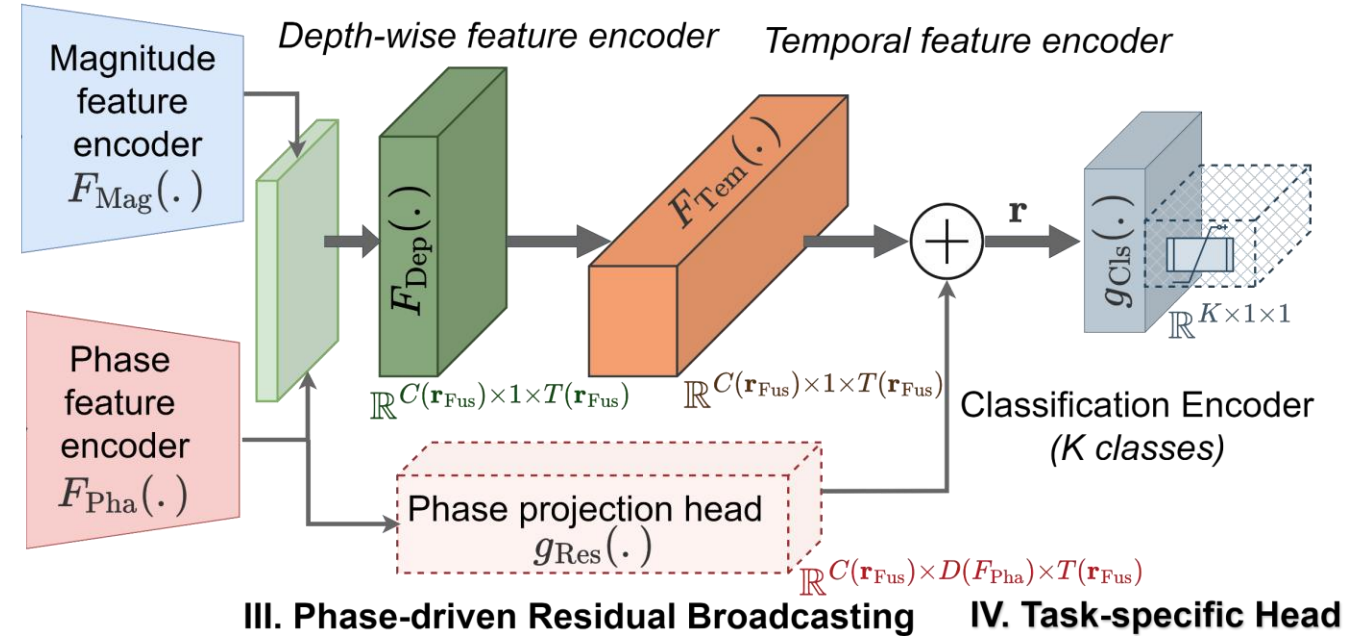
I. Phase-based Augmentation II. Separate Magnitude-Phase Encoding



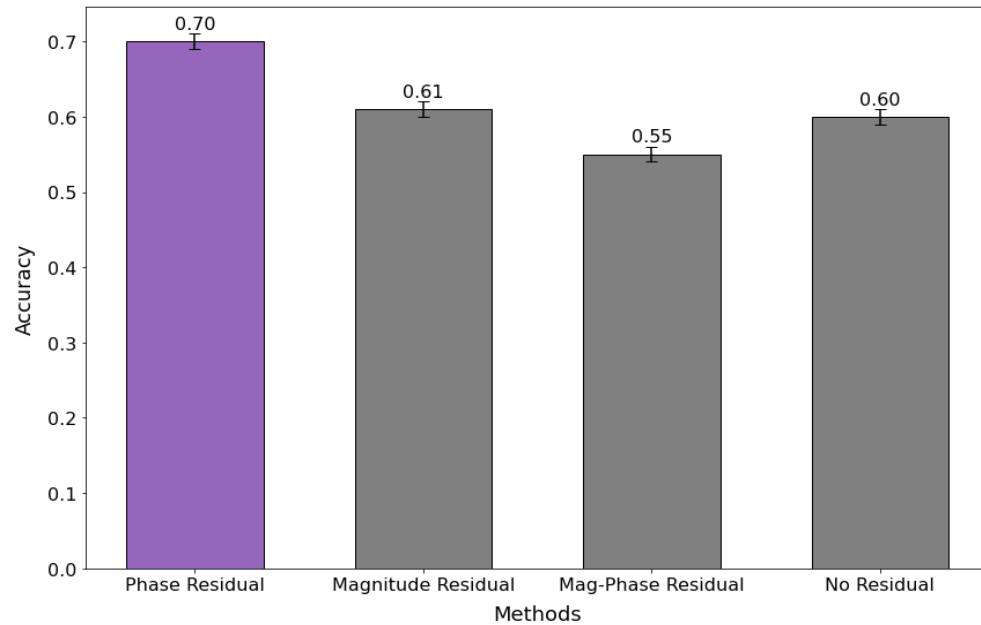
Intuition : Phase and Magnitude contain task and nonstationarity related information distinctly, treating them as separate modalities can help in appropriately modeling them.

3. Phase Residual-Broadcasting Network

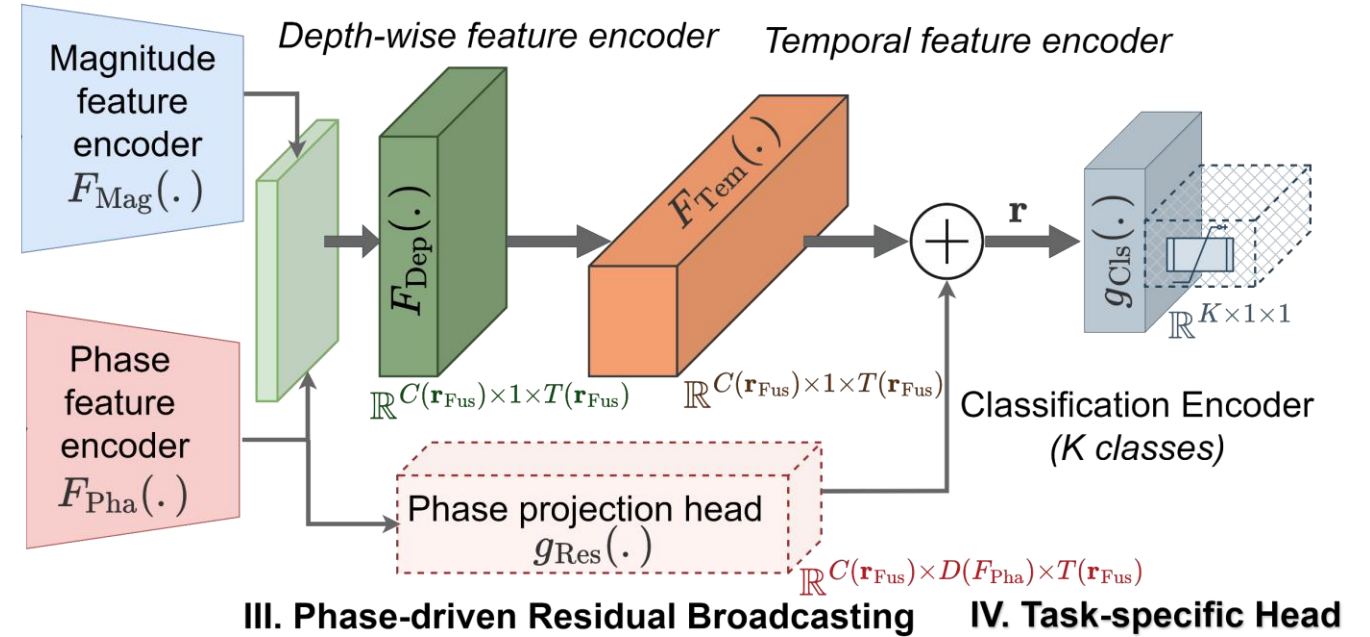
- Residual-Broadcasting Network
 - Collapse to 1D then broadcast using Phase-embeddings to 2D.
- Use Phase-embedding as a residual.



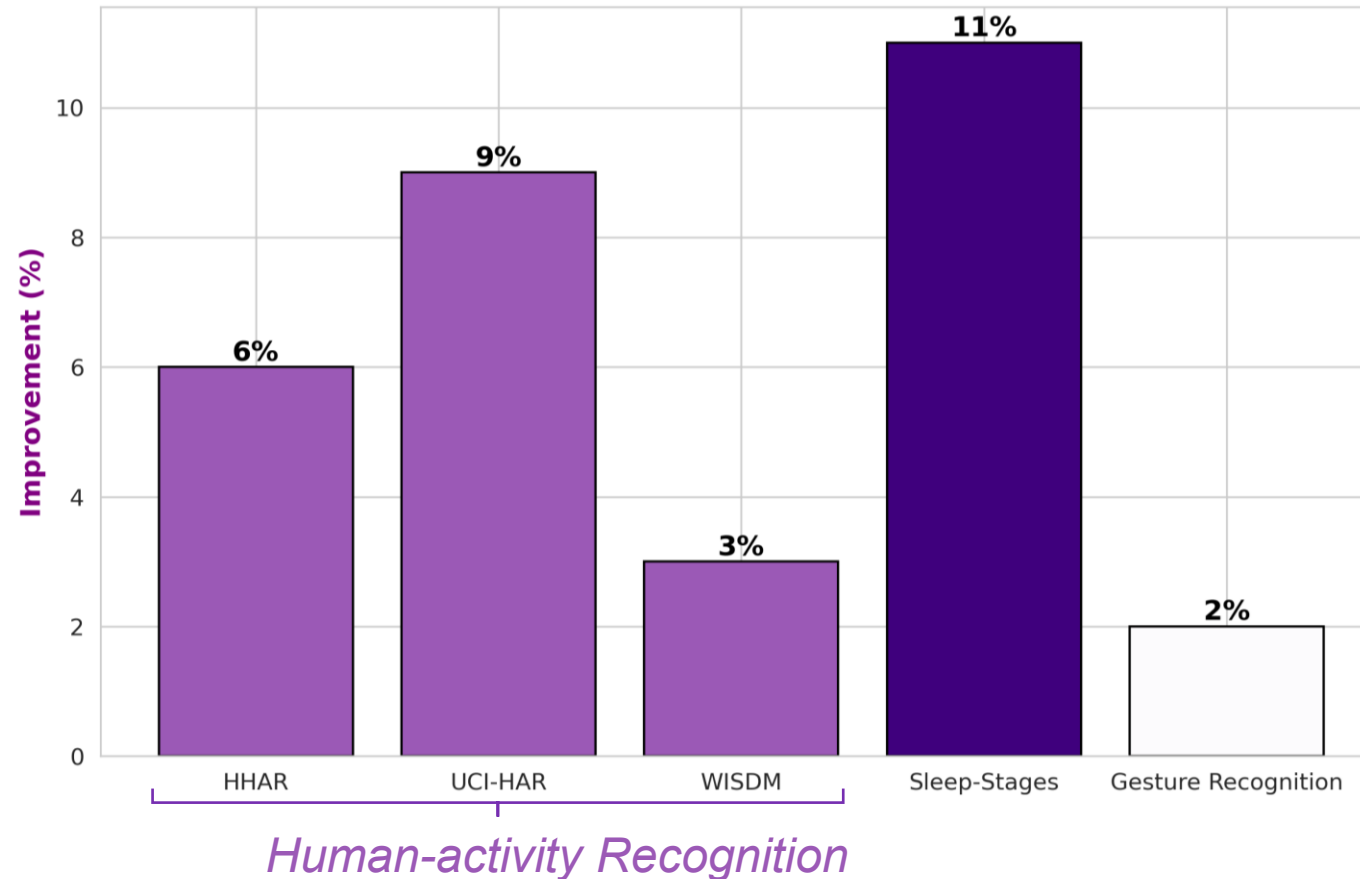
3. Phase Residual-Broadcasting Network



Intuition : Helps in implicit regularization.



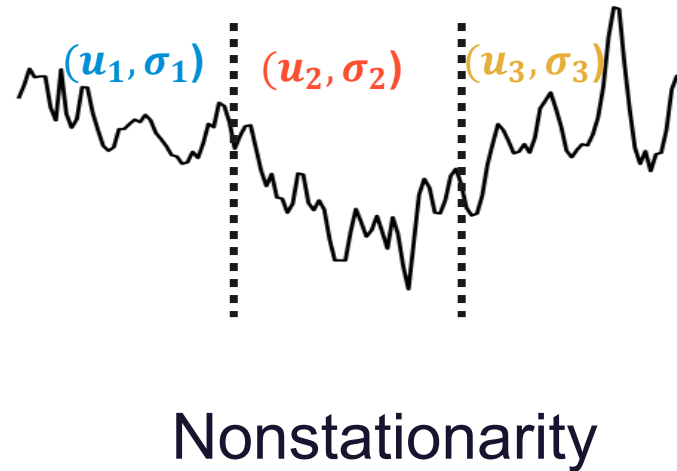
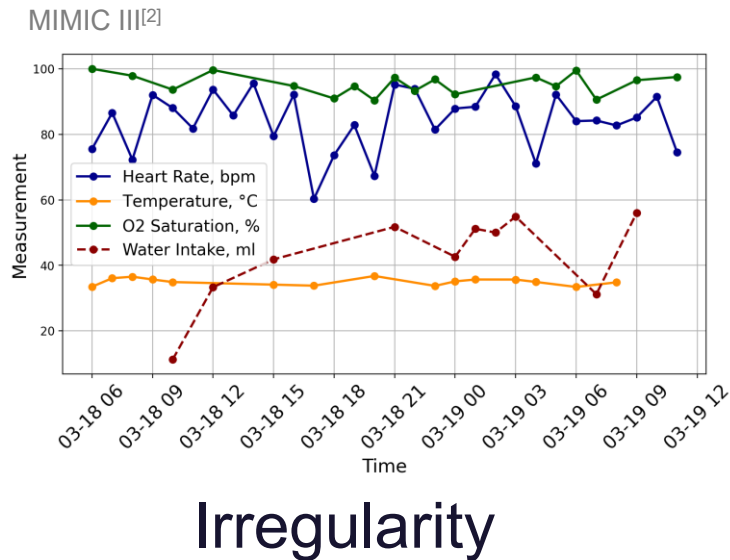
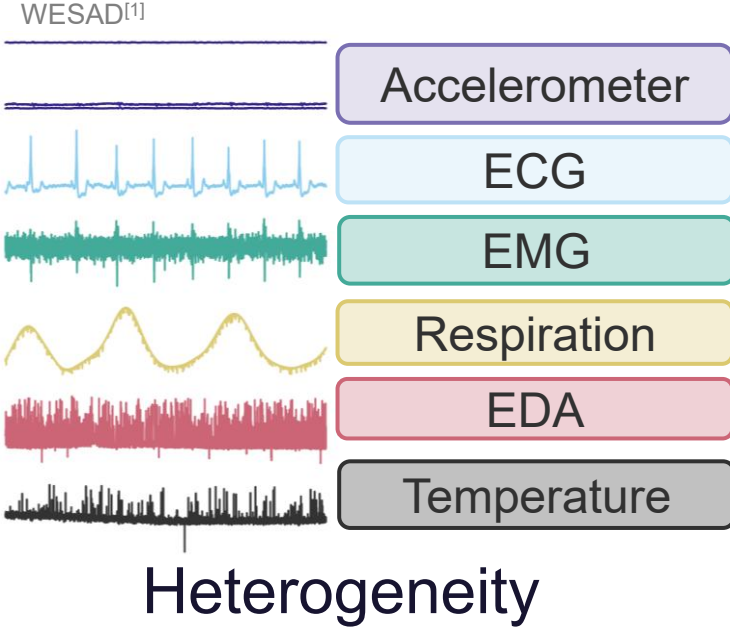
Key Result : PhASER improves generalization across applications.



Summary

- MAESTRO: Multimodal handling of time series using symbolic tokenization and a Mixture-of-Experts framework for **robustness against arbitrary sensor missingness**.
- PhASER: Anchoring design in phase information through augmentation, separate feature encoding, and residual broadcasting to enable **generalizable representations for nonstationary time-series** classification.

Challenges in Time Series Data Analyses



Addressing the Challenges in Time Series Data Analyses

Heterogeneity

P.Mohapatra et.al, Can LLMs Understand Unvoiced Speech? Exploring EMG-to-Text Conversion with LLMs, **ACL 2025**

P.Mohapatra et.al, MAESTRO : Adaptive Sparse Attention and Robust Learning for Multimodal Dynamic Time Series, **Neurips 2025** (*Spotlight*)

Missingness

P.Mohapatra et.al, Missingness-resilient video-enhanced multimodal disfluency detection, **Interspeech 2024**

P.Mohapatra et.al, Person identification with wearable sensing using missing feature encoding and multi-stage modality fusion, **ICASSP 2023** (*top performer in Signal Processing Grand Challenge*)

Distribution Shift

P.Mohapatra et.al, Phase-driven domain generalizable learning for nonstationary time series, **TMLR 2025**

Subjectivity in labels

(Human-centered applications)

P.Mohapatra et.al, Wearable network for multilevel physical fatigue prediction in manufacturing workers, **PNAS Nexus 2024**

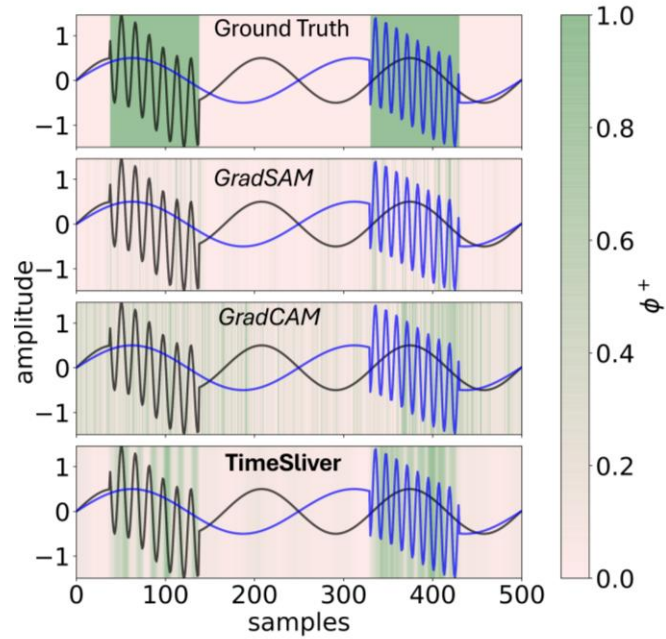
P.Mohapatra et.al, Effect of attention and self-supervised speech embeddings on non-semantic speech tasks, **ACM Multimedia 2023** (*top performer in ComPaRe challenge*)

Data-Efficient Learning

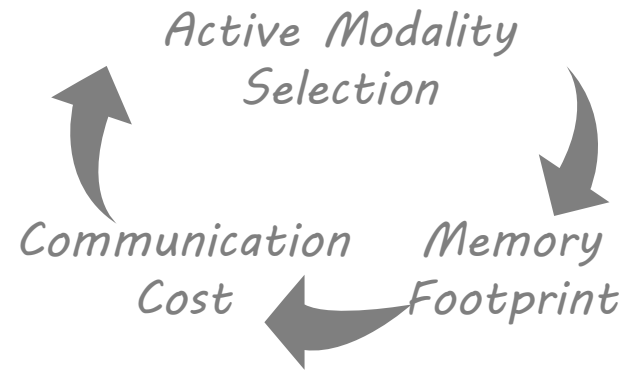
P.Mohapatra et.al, Efficient stuttering event detection using siamese networks, **ICASSP 2023**

P.Mohapatra et.al, Speech disfluency detection with contextual representation and data distillation, **IASA@Mobisys 2022**

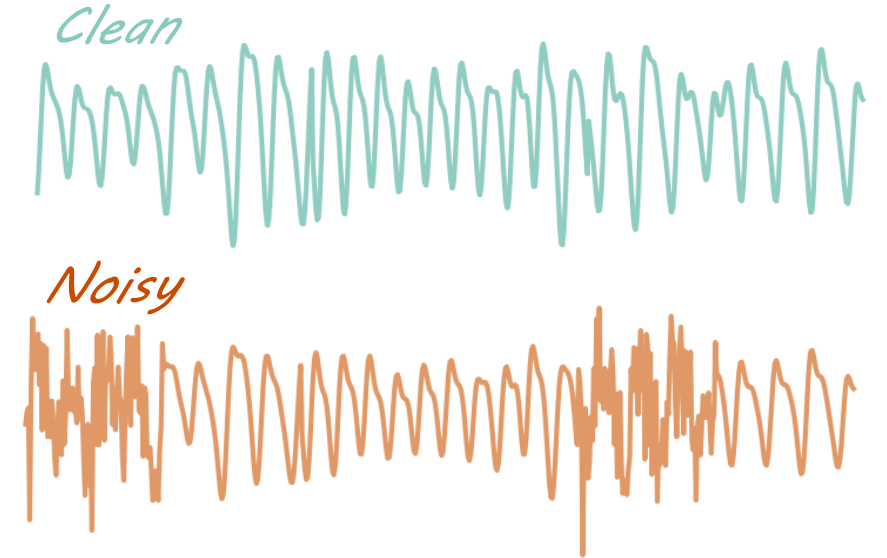
Ongoing Efforts and Future works



Interpretability

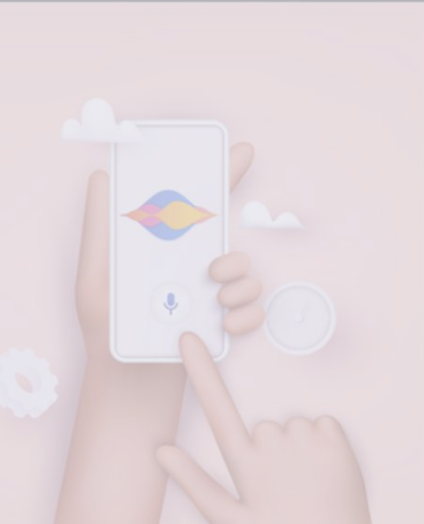


System Efficiency



Noisy Data Streams

...and many more ☺



Thank You.

Payal Mohapatra

Designing technical enablers to improve analytics frameworks
for real world time-series applications.

Connect with me here!



Learn more about my work here!

